

Attitudes toward Immigrants, Information Shocks, and Hate Crime*

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Abstract

We examine how national political shocks can paradoxically amplify hate crimes more in liberal than in already xenophobic communities. Focusing on the surprise UKIP victory in EU elections and Brexit referendum, we hypothesize that because of heterogeneous priors, people in liberal areas experienced stronger belief shock from nationwide xenophobia revealed by these events. When xenophobic minority in liberal communities realized that their attitudes align with national sentiment, they updated beliefs about social acceptability of xenophobia, resulting in larger surges in hate crimes in liberal communities. Using difference-in-differences, we show significant post-event surges in hate crimes in communities with more liberal attitudes toward immigrants. These results hold after accounting for economic and cultural determinants of UKIP and Brexit support. Subsequent analyses show stronger belief shocks among xenophobic individuals in liberal areas, which map onto larger post-event surges in hate crimes. Our findings underscore how heterogeneous priors and strategic complementarity shape hate crime.

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*The author names were randomized using the AEA random author tool. The confirmation code is EibU4ZXnVGZZ.

I. Introduction

There is a growing concern over surge in populism along with racial and religiously motivated hate crimes in democratic societies (Guriev and Papaioannou, 2022). However, our understanding of mechanisms underlying this surge is still limited. In this paper, we investigate whether and how information shocks from *nationwide* political events can lead to surges in hate crime. We leverage a nationally representative dataset to show that when there are heterogeneous priors and political events reveal countrywide support for xenophobia, then strategic complementarity between national and local behaviors can give rise to hate crimes not in xenophobic but *liberal* communities.

How can we explain this counterintuitive result? There is evidence that information shocks revealing previously unknown support for xenophobia can dispel pluralistic ignorance (Lohmann, 1994; Kuran, 1991). Bursztyn, Egorov and Fiorin (2020) show that such information shocks can trigger rapid shifts in social norms, making xenophobic behavior more acceptable. However, this finding cannot explain our result. One plausible mechanism is that individuals hold heterogeneous priors and respond to new information only if it differs from their priors (see Cantoni et al., 2019). When such priors are not randomly distributed but embedded in a community, we get two insights into who experiences a belief shock and for whom these are pivotal for hate crimes. First, individuals form beliefs about the prevalence of nationwide xenophobia by extrapolating from their local environment. This means that, individuals residing in liberal communities tend to believe that the nation is also liberal, whereas individuals residing in xenophobic communities expect xenophobia to be widespread. Second, when a political event publicly reveals nationwide support for xenophobia, it contradicts the priors of individuals residing in liberal areas. The xenophobic minority in these communities updates its beliefs and shifts behavior from abstaining to engaging in hate crimes, as national normalization of xenophobia now lowers their social costs of engaging in such acts. In contrast, the revelation confirms the priors of individuals residing in xenophobic areas, resulting in little changes in hate crimes.

We test this hypothesis using the context of the United Kingdom, which experienced two prominent and unexpected information shocks that revealed previously hidden anti-immigrant sentiment at the national level. The first information shock was the unexpected victory of UK Independence Party (UKIP) in the European Parliament election, 2014. UKIP not only increased its vote share by over 70 percent, but it was also the first time that a party other than Labour or Conservative had won any election since 1910. The Guardian reported it as a “political earthquake”. The second information shock was the unexpected approval of the Brexit referendum in 2016, which resulted in the UK leaving the European Union. The BBC described the outcome as a “huge surprise.” In both events, there was an explicit use of anti-immigrant rhetoric followed by a large surge in

hate crime in post-event periods.

We focus on hate crimes motivated by race, ethnicity, and religion. These crimes are subject to stricter sentencing than the equivalent crime without the motivation.¹ We collect quarterly data on hate crime at the level of local units called Community Safety Partnerships (CSP), which are empowered to formulate and implement strategies to tackle local crime. Our sample includes 97% of the CSPs in England and Wales, and spans 18 years (2002-2019).²

We capture xenophobia as ‘attitudes toward immigrants’ using survey data from two waves of British Election Study (BES) that were conducted before the events. In the survey, respondents are asked to indicate their support for immigrants on a seven-point scale, such that higher number implies stronger support. We focus on two questions “immigrants are good or bad for the British economy” and “immigrants undermine or enrich Britain’s cultural life” to capture economic and cultural dimensions. We refer to CSPs with stronger support toward immigrants as *liberal* and those with weaker support as *xenophobic*.

We start by providing two pieces of empirical evidence in support of our conceptual framework. First, we argued that people use local xenophobic sentiment to project nationwide xenophobia, thereby assuming that people are aware of the extent of xenophobia in their own CSP. To this end, we show a strong positive association between their belief and actual local support for xenophobia. Second, we show that both liberal and xenophobic people in more liberal CSPs experienced a larger belief shock that their counterparts in mostly xenophobic CSPs.

We proceed by examining the effect of attitudes toward immigrants on hate crime in the post-event periods. We test this using a difference-in-differences design. In our study, all CSPs experienced UKIP and Brexit event, but there is a heterogeneous response depending on attitudes toward immigrants.³ Our specification includes CSP and quarter fixed effects, as well as region specific time trends. This allows us to account for unobserved time-invariant differences across CSPs and common shocks that affect all CSPs and regions similarly in a given quarter. In addition, we control for CSP level covariates that change over time including population size, number of EU immigrants, and number of non-EU immigrants. Conditional on fixed effects and controls, the identifying assumption is that absent UKIP and Brexit events hate crimes would have evolved similarly in CSPs with different attitudes toward immigrants.⁴ The event study plots clearly show no pre-trends

¹Examples include racist graffiti sprayed on someone’s home or property, verbal abuse in the form of using slurs, and physical assault accompanied by racist language.

²We could not get locally disaggregated data on hate crime in Scotland and Northern Ireland, so we exclude these from our study. We exclude data after 2019 because of confounding with COVID-19.

³See Korovkin and Makarin (2023) who use a similar strategy to study the effect of Russia-Ukraine conflict on trade patterns depending on the share of Ethnic Russians in the districts of Ukraine.

⁴Though UKIP and Brexit events are sequenced over time, in each event every CSP is treated. So, this is not a case of staggered treatment in which different units are treated at different points in time.

in hate crime, lending credibility to our identifying assumption.

We find a strong positive association between attitudes towards immigrants and surges in hate crimes in the post-event periods, which are statistically significant at the 1% level. Our estimates suggest that one standard deviation increase in attitudes is associated with a rise in hate crime by 0.11-0.16 standard deviations. The increase in hate crime does not diminish over time. This means our findings cannot be due to busloads of xenophobic people traveling to liberal areas to commit hate crimes, as this can explain a surge in hate crime in only one quarter but not a sustained effect over 16 quarters.

We foresee two major empirical challenges. First, as Guriev and Papaioannou (2022) note, economic, cultural, and social factors can gradually build-up support for populist movements and xenophobia. In the context of the UK, Fetzer (2019) shows that austerity led to a rise in support for UKIP and eventually Brexit. However, the possibility that factors that gave rise to the events also account for surges in hate crime in post-event periods is unlikely to be of concern. This is because these factors are typically negatively associated with attitude toward immigrants and are therefore unlikely to explain our counterintuitive results. Nevertheless, we take steps to separate the effect of information revelation by the events from the effect of factors that give rise to the events. While conducting robustness checks, we show that our results hold when we include post-event interactions with austerity, education deprivation, generalized trust, and trust in government. A related concern is that poor economic conditions arising from the events themselves led to surges in hate crime. However, while there is an immediate rise in hate crime in the post-event periods, economic changes appeared with a lag of several years (see for instance, Bakker et al., 2022).

Second, our results could be capturing changes in recording of hate crimes especially due to increased sensitivity among the police or directives to record hate crimes more accurately. We show that our results are robust to controlling for either police-force specific time trends or an interaction between the post-event and police-force indicators. A related concern is that the frequency of hate crimes is the same, but reporting increased in the post-event periods. Addressing this concern fully is very difficult, but we take two steps to assuage this concern. We start by using data from Crime Survey of England and Wales on reporting rates of different types of crimes that also have hate counterparts. Using an event study we show that there is no change in reporting rates in the post-event periods. CSPs vary widely in their composition of types of hate crime, which allows us to adjust our measure of hate crime using reporting rates specific to the type of crime. All of results hold when we use this adjusted measure. Finally, we show our results are robust to considering one at a time different type of hate crime, which differ widely in their reporting rate.

Finally, we construct falsification tests using data on crimes like murder, burglary, and driving violations. We find no changes in hate crime by attitude toward immigrants

in the post-event periods, which is expected as these crimes are unlikely to arise from information shocks generated by UKIP and Brexit events.

We now turn to understanding the mechanism underlining the above results. In line with our conceptual framework, we show that using both event studies and difference-in-differences specification that CSPs with stronger belief shocks experienced a significantly higher surge in hate crime in the post-event periods by 0.06-0.11 standard deviations.

Since it is not possible to get data on hate crime at the individual level, we carry out a proof-of-concept survey with close to 1500 participants. In the survey, we ask participants on their willingness to express views over immigrants. We find that xenophobic people living in more liberal areas are significantly more likely to express their views in the post-event period.

Our paper contributes to the literature on rise in populism and hate crimes (Guriev and Papaioannou, 2022). Several studies show that areas with historically persistent xenophobia are more likely to experience surges in hate crimes (Voigtländer and Voth, 2012), especially in response to persuasion campaigns and coordination facilitated by radio (Adena et al., 2015), and social media (Bursztyn et al., 2019). A closely related paper is by Bursztyn, Egorov and Fiorin (2020) who show that when information shocks reveal that the local population is xenophobic, individuals respond with higher levels of xenophobic actions. A common theme across these impressive studies is that these surges occur in areas that were already xenophobic. Using nationally representative data, we show that allowing for heterogeneous priors in belief updating implies that information shocks can lead to a counterintuitive increase in hate crime in liberal rather than xenophobic areas. Our paper also distinguishes itself from Cantoni et al. (2019) who study the role of heterogeneous priors in the context of participation in anti-authoritarian protests in Hong Kong. By leveraging how priors are formed in the context of a community, we can predict where and for whom change in behavior occurs.

Our paper also contributes to the emerging literature on hate and toxicity on social media (Bursztyn et al., 2019; Jiménez-Durán, 2023; Jiménez Durán, Müller and Schwarz, 2024; Kalra, 2024; Beknazar-Yuzbashev et al., 2025). These studies show that exposure to extremist or hateful content can reveal latent preferences, reduce the perceived social costs of expressing xenophobia, and facilitate coordination among like-minded individuals, often leading to increases in hateful speech. Fages and Martínez-Pozo (2026) document a temporary rise in sexist and homophobic speech on social media in the aftermath of Bolsonaro’s election in areas where he received weaker vote share. However, this study does not posit any mechanisms underlying this result. Our paper complements this literature by focusing on offline hate crime and by identifying national political events—rather than platform-level or algorithmic shocks—as a source of information. Moreover, by emphasizing heterogeneous priors formed from local environments, we show that information shocks can disproportionately activate xenophobic minorities in otherwise tolerant areas.

II. Conceptual Framework

In this section we first propose a theoretical model of how individuals choose their behavior towards immigrants based on their personal preferences and their beliefs about the behavior of other individuals. We then discuss how the insights we obtain can be applied to the specific case of hate crime.

Background.— We consider a country of measure 1 with a continuum of individuals who are divided into geographical districts of equal size (for simplicity). All individuals move simultaneously. Each individual i in district d selects an *action* $a_i \in \mathbb{R}$ to maximize expected utility. This depends on (i) how closely the action matches a preference parameter $\alpha_i \in \mathbb{R}$ reflecting the individual’s intrinsic preferences, and (ii) how closely the action conforms to a reference action $\bar{a}_R^d$, defined as $\bar{a}_R^d \equiv \lambda \bar{a} + (1 - \lambda) \bar{a}^d$, where $\bar{a}^d$ and $\bar{a}$ represent mean actions in district d and countrywide, respectively, and $0 < \lambda \leq 1$.

The reference action $\bar{a}_R^d$ captures the desire to conform to local as well as countrywide behavior, with the relative importance of the latter being parameterized by λ . These conformity concerns may originate from a desire to fit in, to behave in a socially acceptable way – where social acceptability is defined by a combination of local and countrywide components – or to behave in a way that is consistent with one’s identity – as e.g. in George A Akerlof and Rachel E Kranton (2000); Roland Bénabou and Jean Tirole (2011).⁵ In this interpretation, λ captures the extent to which individuals identify with the country as opposed to their local area.

Individual preferences.— The individual preference parameter is given by the sum of a district-specific component P^d , and an individual-specific component, ε_i . The district-specific component is in turn equal to the sum of a mean preference parameter μ and a component that captures district-specific characteristics, e^d . To sum up, therefore, the preference parameter α_i^d of an individual i in district d is

$$\alpha_i^d = \underbrace{\mu + e^d}_{= P^d} + \varepsilon_i \quad (1)$$

where μ represents (unobservable) countrywide mean preferences, $e^d \sim N(0, 1)$ is a district-specific component and $\varepsilon_i \sim N(0, \sigma)$ is an individual-specific component.

⁵We follow George A Akerlof (1980); Moti Michaeli and Daniel Spiro (2017); P Grout, S Mittraille and S Sonderegger (2015) and Cristina Bicchieri, Eugen Dimant and Silvia Sonderegger (2023) in adopting a consequentialist approach, in the sense that an individual’s utility depends on how the individual behavior relates to the reference behavior. Another branch of the literature, such as B Douglas Bernheim (1994); Roland Bénabou and Jean Tirole (2011) or Fabrizio Adriani and Silvia Sonderegger (2019), focuses instead on the case where behavior generates social or psychological utility only to the extent to which it reveals information about an individual’s underlying type.

Information.— Consistent with the empirical results provided in Figure 3 below, people have a good grasp of preferences at the local level. Formally, this is captured by letting them observe P^d , their local mean preferences.⁶ In addition, all individuals observe a public signal $\bar{\mu} = \mu + \xi$ where $\xi \sim N(0, \Theta)$. We can think about $\bar{\mu}$ as the countrywide, publicly available information about μ . This may come from shared history and cultural traits, as well as information-revelation events at country level, such as elections or referenda.

Beliefs.— The beliefs about μ in district d are a weighted average of the information available to individuals, namely local preferences P^d and the public information $\bar{\mu}$,

$$E(\mu \mid P^d, \bar{\mu}) = \frac{\Theta}{1 + \Theta} P^d + \frac{1}{1 + \Theta} \bar{\mu}. \quad (2)$$

This highlights that beliefs are *heterogeneous* across districts. Intuitively, when forming beliefs about countrywide preferences, people partially extrapolate from local preferences.⁷ Individuals living in liberal areas tend to believe that the country as a whole is liberal, while those living in xenophobic areas tend to hold the opposite belief.

Utility.— The utility of any individual i in district d with personal preference α_i is composed of two loss functions, capturing the psychological cost of deviating from one's own personal preferences as well as the social and psychological cost of deviating from the reference action. Formally,

$$u_i = -\theta (a_i - \alpha_i)^2 - (1 - \theta) (a_i - \bar{a}_R^d)^2. \quad (3)$$

where $\theta \in (0, 1)$ captures the concern for aligning behavior with personal preferences relative to conforming with the reference behavior and, as mentioned, $\bar{a}_R^d \equiv \lambda \bar{a} + (1 - \lambda) \bar{a}^d$.

II.A. Equilibrium Characterization

Our equilibrium concept is Perfect Bayesian Equilibrium. Each individual i in district d chooses a_i to maximize the expectation of (3), where expectations are consistent with equilibrium behavior. Differentiating the objective function and rearranging delivers,

$$a_i = \theta \alpha_i + (1 - \theta) E(\bar{a}_R^d \mid P^d, \bar{\mu}) \quad (4)$$

⁶The analysis can be easily extended to the case where P^d is not fully observable, at the cost of additional analytical complexity.

⁷This is superficially reminiscent of the false consensus effect, a well known psychological bias that refers to the tendency of people to overestimate the extent to which their preferences (in this case, preferences in their local areas) are typical of those of others (here, other areas). However, in our analysis, using preferences in one's local area to predict countrywide preferences is not due to a bias but is actually consistent with Bayesian updating – see also Christoph Vanberg (2019) and Fabrizio Adriani and Silvia Sonderegger (2015) for illustrations of this general point.

where $E(\bar{a}_R^d | P^d, \bar{\mu}) = \lambda E(\bar{a} | P^d, \bar{\mu}) + (1 - \lambda) E(\bar{a}^d | P^d, \bar{\mu})$.

Expression (4) shows that each individual's action is a weighted average of own preferences and expectations about the reference action. Through this mechanism, the model generates strategic complementarities between individual behavior and mean behavior at both the local and country levels.⁸

Proposition 1 *In the unique linear symmetric equilibrium of the game, we have*

$$a_i = \theta \alpha_i + \gamma P^d + (1 - \theta - \gamma) \bar{\mu} \quad (5)$$

where $\gamma \equiv \frac{\theta(1-\theta)(\Theta+1-\lambda)}{\theta+\lambda(1-\theta)+\theta\Theta} < 1 - \theta$ increasing in Θ .

Proof: The proof is provided in the Appendix.

Corollary 1 (i) *Mean countrywide behavior is $\bar{a} = (\theta + \gamma) \mu + (1 - \theta - \gamma) \bar{\mu}$.*

Individuals form their expectations about countrywide behavior from their beliefs about countrywide preferences: $E(\bar{a} | P^d, \bar{\mu}) = (\theta + \gamma) E(\mu | P^d, \bar{\mu}) + (1 - \theta - \gamma) \bar{\mu}$, where $E(\mu | P^d, \bar{\mu})$ is given by (2). Substituting for this and for $\bar{a}^d$ in expression (4) we obtain (5).

Corollary 1 (ii) *Mean behavior in district d is $\bar{a}^d = (\theta + \gamma) P^d + (1 - \theta - \gamma) \bar{\mu}$.*

The term $(\theta + \gamma) P^d$ in $\bar{a}^d$ reflects behavioral heterogeneity across districts. The first component, θ , captures people's desire to conform to their own (district-specific) preferences, while the second component, γ , captures cross-district heterogeneity in beliefs about μ .

II.B. Effect of Information Shock on Behavior

We now consider the effect of an information shock that releases new public information about μ , the mean countrywide preferences. Let $\bar{\mu}_0 \sim N(\mu, \Theta_0)$ be the pre-shock publicly observable signal about μ , so that $E(\mu | \bar{\mu}_0) = \bar{\mu}_0$, $Var(\mu | \bar{\mu}_0) = \Theta_0 > 0$. The post-shock public information is denoted as $\bar{\mu}_1$, with $E(\mu | \bar{\mu}_1) = \bar{\mu}_1$, $Var(\mu | \bar{\mu}_1) = \Theta_1 < \Theta_0$.⁹ Without loss of generality, we let higher values of a (resp., α) capture more positive behavior (resp., preferences) towards immigrants. We consider the case $\bar{\mu}_1 < \bar{\mu}_0$, i.e. the new information indicates that countrywide preferences are less positive than previously thought.

We start by assessing cross-district differences in the effect of the information shock on beliefs. As shown in (14), beliefs about μ are heterogeneous across districts. This implies that the change in beliefs generated by the shock is also heterogeneous.

⁸For analytical convenience, we consider a continuum of districts.

⁹Formally, let the additional signal released by the information shock be $\bar{\mu}_2 = \mu + \xi_2$ where $\xi_2 \sim N(0, \Theta_2)$. By Bayesian updating, the posterior belief conditional on both $\bar{\mu}_0$ and $\bar{\mu}_2$ is that μ is normally distributed around $\bar{\mu}_1 \equiv \frac{\Theta_2 \bar{\mu}_0 + \Theta_0 \bar{\mu}_2}{\Theta_0 + \Theta_2}$ with variance $\Theta_1 \equiv \frac{\Theta_0 \Theta_2}{\Theta_0 + \Theta_2}$.

Lemma 1 *The difference in beliefs in district d following the shock is*

$$E(\mu \mid P^d, \bar{\mu}_0) - E(\mu \mid P^d, \bar{\mu}_1) = \frac{1}{1 + \Theta_0} \bar{\mu}_0 - \frac{1}{1 + \Theta_1} \bar{\mu}_1 + \frac{\Theta_0 - \Theta_1}{(1 + \Theta_1)(1 + \Theta_0)} P^d, \quad (6)$$

increasing in P^d .

Proof: Follows straightforwardly from Bayesian updating.

Intuitively, information indicating that the country is xenophobic is less surprising to individuals living in xenophobic areas, because their prior beliefs already suggested that the country may be xenophobic.

The next result highlights the implications of Lemma 1 for the cross-district change in behavior triggered by the shock. Since the new information points to more xenophobia at country level than previously thought, behavior generally becomes more xenophobic, due to strategic complementarity between individual and countrywide behavior.¹⁰ However, the *size* of this shift differs across districts. Denote as γ_t , $t \in \{0, 1\}$, the value of γ when $\Theta = \Theta_t$. Note that, since $\Theta_0 > \Theta_1$, we have $\gamma_0 > \gamma_1$.

Corollary 2 *The change in mean behavior in district d following the shock is*

$$\bar{a}_0^d - \bar{a}_1^d = (1 - \theta - \gamma_0) \bar{\mu}_0 - (1 - \theta - \gamma_1) \bar{\mu}_1 + (\gamma_0 - \gamma_1) P^d, \quad (7)$$

increasing in P^d .

In words, the model predicts that more xenophobic areas experience a smaller behavioral response to the information shock. That's because individuals in xenophobic areas are less surprised by the new information than those in liberal areas and therefore adjust their behavior to a lesser extent.

II.C. Implications for Hate Crime

In this section we describe how the setup of the previous section maps into hate crime. We think of hate crime as an extreme expression of overt dislike towards immigrants (or, more generally, people from a minority background). The simplest way to capture this is to consider a threshold $-\mathbf{a} < 0$, such that, when behavior is below this threshold, it is classified as hate crime. Recall that, in our setup, utility is

$$u_i = -\theta (a_i - \alpha_i)^2 - (1 - \theta) (a_i - \bar{a}_R^d)^2 \quad (8)$$

$$= 2a_i [\theta \alpha_i + (1 - \theta) \bar{a}_R^d] - a_i^2 + K \quad (9)$$

¹⁰We are implicitly focusing on the empirically relevant case where $E(\mu \mid P^d, \bar{\mu}_1) < E(\mu \mid P^d, \bar{\mu}_0)$ and $\bar{a}_1^d < \bar{a}_0^d$.

where $K \equiv -\theta\alpha_i^2 - (1 - \theta) (\bar{a}_R^d)^2$. This makes clear that, although we have expressed utility as a combination of two loss functions, we can equivalently think of it as the result of the trade-off between the net psychological and social return from selecting action a_i — given by $2a_i [\theta\alpha_i + (1 - \theta) \bar{a}_R^d]$ — and the expected material cost arising from the legal sanctions associated with a_i — given by a_i^2 . A possible micro-foundation for the expected cost of legal sanctions is the following. All extreme behaviors —whether strongly anti-minority ($a_i < -\mathbf{a}$) or strongly anti-majority ($a_i > \mathbf{a}$) — are considered crimes, with the former categorized as hate crimes, and people are uncertain about the exact value of $\mathbf{a}$.¹¹

The effect of information shock on hate crime.— Let $\tilde{a}_t^d \equiv (-\mathbf{a} - \bar{a}_t^d)/\theta\sqrt{\sigma}$ where $\bar{a}_t^d = (\theta + \gamma_t)P^d + (1 - \theta - \gamma_t)\bar{\mu}_t$, $t \in \{0, 1\}$, and denote as $F(\cdot)$ the cdf of the standard normal distribution. In district d , the change in hate crime generated by the shock is equal to $F(\tilde{a}_1^d) - F(\tilde{a}_0^d)$. Differentiating with respect to P^d we obtain

$$\frac{1}{\theta\sqrt{\sigma}} [f(\tilde{a}_0^d) (\gamma_0 - \gamma_1) + (\theta + \gamma_1) (f(\tilde{a}_0^d) - f(\tilde{a}_1^d))] \quad (10)$$

where, as already observed, $\gamma_0 > \gamma_1$. There are two effects at play. The first is that, as highlighted in Corollary 2, the behavioral shift induced by the shock is less pronounced in more xenophobic areas. In (10), this is captured by $f(\tilde{a}_0^d) (\gamma_0 - \gamma_1) > 0$. The second effect relates to the probability density of the preference parameters of those people who are pushed below the hate crime threshold by the information shock. In the Appendix, we prove that if the within-district variance of individual preferences sufficiently large, expression (10) is always positive. The increase in hate crime following the shock is thus unambiguously larger in more liberal areas.

III. Field Setting and Data

The UK provides an ideal setting for our study because it witnessed two prominent nationwide information shocks that exposed previously hidden anti-immigrant attitudes at the national level. We provide an overview of these events below, followed by data on hate crime and attitudes. Finally, we present evidence in support of our conceptual framework by showing that local environments shape belief formation and individuals from liberal CSPs experienced larger belief shocks.

¹¹To fix ideas, consider $a_i < 0$ (the case $a_i > 0$ is analogous). Suppose that, from the public's perspective, $\mathbf{a} \sim U[0, \bar{\mathbf{a}}]$, where $\bar{\mathbf{a}}$ is “large”, and let the cost of being convicted for a hate crime be $b(-a_i - \mathbf{a})$. When $b = 2\bar{\mathbf{a}}$, the expected cost associated with a_i is $b\Pr(-\mathbf{a} > a_i) E(-\mathbf{a} - a_i \mid -\mathbf{a} > a_i) = a_i^2$ for all $a_i \geq -\bar{\mathbf{a}}$.

III.A. National Political Events

UKIP’s Election.— The first information shock is the UK Independence Party’s (UKIP) unexpected victory in the European Parliament elections held in May 2014, in which UKIP increased its vote share by more than 70 percent relative to the previous election, from 15.6% in 2004 to 26.6% in 2014. It was the first time since 1910 that a party other than Labour or Conservative had won the largest number of seats in any election. A leading newspaper in the UK, the Guardian, reported this as a “political earthquake.”¹² UKIP was the only party that explicitly advocated for leaving the European Union and used xenophobic rhetoric, so its surprise victory signaled that anti-immigrant attitudes were more widespread among the population than previously expected.

Brexit Referendum.— The second information shock is the Brexit referendum that was held in June 2016, a vote that resulted in the UK eventually leaving the European Union (EU). The referendum outcome was described by the BBC as “a huge surprise.”¹³ In fact, on voting day, British bookmakers placed the likelihood of a Leave victory to be between 13% and 17% — thus vastly underestimating the outcome. The Leave campaign made explicit use of anti-immigrant rhetoric, framing immigration as a central issue in its appeal to voters. In the aftermath of Brexit, hate crimes in Britain hit record high.¹⁴

Note that voter turnout in the Brexit referendum was twice as large as in the UKIP election. This means that the information it released about the distribution of attitudes towards immigrants in the country as a whole was more accurate. This implies that Brexit event could have generated a belief shock over and above UKIP event. We provide evidence on this in Section VI.

III.B. Data

In the UK, crime data are recorded by the 43 territorial police forces. Each police force typically oversees several Community Safety Partnership (CSP) areas within its jurisdiction. The CSPs are empowered under the Crime and Disorder Act (1998) to formulate and implement strategies to tackle local crime, disorder and antisocial behavior in their communities. Our unit of analysis is a CSP. We focus on England and Wales, as data from Scotland and Northern Ireland are not available at the CSP level.

We use several different sources to collect CSP level data on hate crime, other crimes, attitudes towards immigrants, and beliefs on the likelihood of UKIP election as well as Brexit. We supplement these analyses with a survey measuring individual level propensity to express personal views towards immigrants following the Brexit referendum. We

¹²Available at <https://www.theguardian.com/politics/2014/may/26/ukip-european-elections-political-earthquake>

¹³Available at <https://www.bbc.com/news/uk-politics-eu-referendum-36616028>

¹⁴Available at <https://www.bbc.co.uk/news/uk-38976087>.

describe the data below, except for data on beliefs and expression of personal views, which we present in section VI.

Hate Crime.— The police force in England and Wales records a criminal offense in the category of hate crime if it is perceived to be motivated by hostility or prejudice towards someone based on their race, ethnicity or religion. These crimes are defined by statute and will typically be subject, if prosecuted, to stricter sentencing than the equivalent crime, absent the racial or religious motivation. The data on hate crime are publicly available from the Office for National Statistics (ONS). It includes violence without injury, public order offenses, criminal damage, and violence with injury.¹⁵

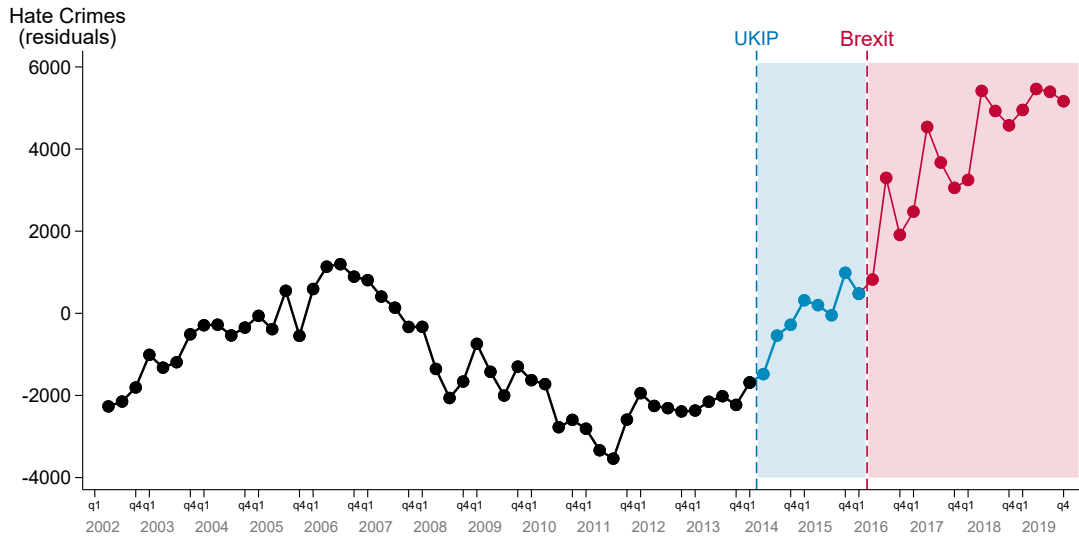


Figure 1: Evolution of Hate Crime over Time (Residuals)

Notes. The figure depicts the evolution of reported hate crimes. The Y-axis displays the residuals of reported hate crimes obtained from a CSP-level regression of the total number of hate crimes on season indicators to capture the seasonal variation in hate crime and an indicator variable that takes the value of 1 during Q3 and Q4 of 2005 and 0 otherwise, to account for the 7/7 London Bombing. The blue solid line corresponds to the post-UKIP period, while the red line corresponds to the post-Brexit period.

The data on hate crime are at the level of a local authority. There are 315 local authorities in England and Wales, which uniquely correspond to 298 CSPs. Five local authorities have multiple CSPs in them, so we assign hate crime in these authorities to the most populous CSP. Our final sample comprises 304 CSPs. We use quarterly data on hate crime for 71 quarters spanning 18 years, from 2002 to 2019.¹⁶ We do not use data from 2020 onwards because of confounding with COVID pandemic. Figure 1 plots total hate crime in England and Wales over time after accounting for an indicator for 7/7

¹⁵Hate crime also includes crimes related to grievous and actual bodily harm. These crimes make up less than 3 percent of all crimes. We exclude these crimes because they do not relate to our mechanism. However, given their small share all of our results hold when we include them.

¹⁶One CSP in Wales, Merthyr Tydfil, has fewer observations because it was merged with Rhondda Cynon Taf CSP by South Wales Police in April 2018. Therefore, hate crimes in this CSP are not reported separately from 2018Q2 onwards.

London bombing and indicators for seasons. It shows large variation. Crucially, there is a steep and steady surge in hate crime post UKIP (shaded in blue) and Brexit (shaded in pink) events. This pattern is not observed in periods prior to the events. Panel A of Table A.1 reports summary statistics on hate crime per CSP per quarter. It ranges between 0 and 616, the average being 31.33 (s.d. 42.78). In the econometric analysis, we use standardized measure of hate crime, with mean 0 and standard deviation of 1.

Attitudes Toward Immigrants.— We collect data on attitudes toward immigrants from the British Election Study (BES), which is an individual level panel survey with approximately 30,000 respondents. We use data on attitudes from panel waves that were conducted just before the UKIP and Brexit events, respectively.

- For the UKIP event, we use data from Wave 1 of the BES, which was conducted between 20 February 2014 and 9 March 2014, two months before the EU Parliament Election.
- For the Brexit event, we focus on Wave 8 of the BES, which was conducted on 11 May 2016, slightly more than a month before the Brexit referendum.

We measure attitudes using responses to two questions: (a) “Do you think immigration is good or bad for Britain’s economy?” The respondents could choose their answer on a scale of 1-7, where 1 implies “bad for the economy” and 7 implies “good for the economy”; and (b) “Do you think that immigration undermines or enriches Britain’s cultural life?”. The respondents could choose their answer on a scale of 1-7, where 1 implies “undermines” and 7 implies “enriches”. The BES data has identifiers for local authority, which we match with CSP using the same procedure that we used for matching hate crime. Since both attitudes are measured on the same scale, we take their average at the CSP level. Our results hold when we consider each attitude separately. Broadly speaking, these attitudes can be considered as capturing important cultural traits like tolerance and respect towards immigrants Tabellini (2010). Panel A of Table A.1 shows that attitudes toward immigrants range from 2.62 to 5.71, the average is 3.636 (s.d. 0.458). In the econometric analysis, we use standardized measure of attitudes with mean 0 and standard deviation of 1.

Given the primacy of attitudes toward immigrants in our analysis, we validate that our measures are not simply capturing noise but have meaningful associations with political outcomes of interest. Table A.2 shows a strong negative and statistically significant association between attitudes toward immigrants and vote share of UKIP in column 1 and support for leave campaign of Brexit in column 2. Furthermore, we show that in Figure 2 that these attitudes are fairly stable and do not decline in response to the events. If at all, they increase slightly in magnitude over time.

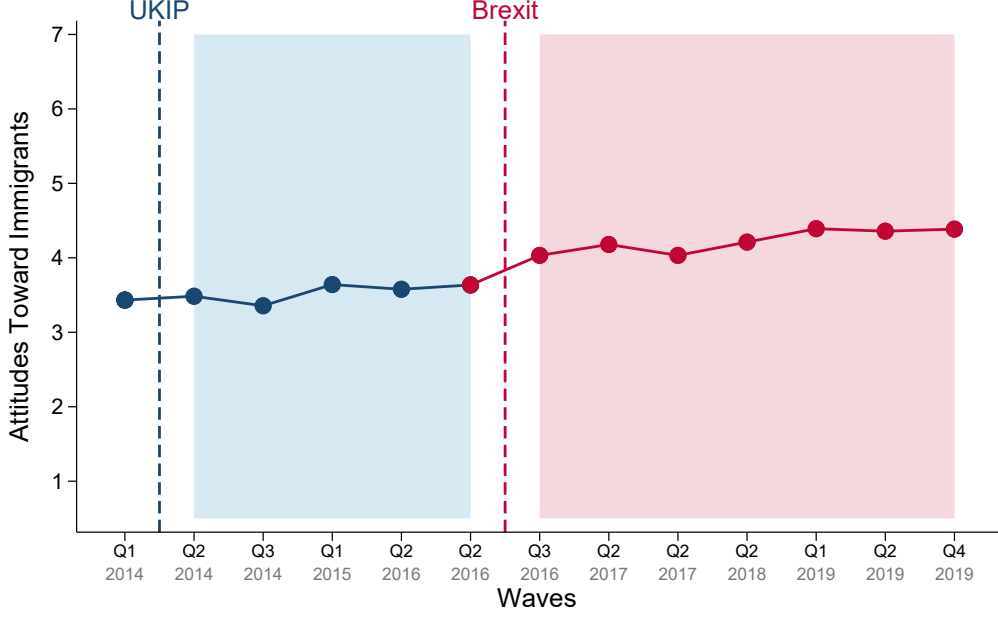


Figure 2: Evolution of Attitudes Toward Immigrants Over Time

Notes. The figure shows the evolution of average attitudes toward immigrants over time. The Y-axis reports attitudes on a 1–7 scale, where 1 indicates very negative and 7 very positive attitudes. The X-axis reports calendar quarters corresponding to survey waves. Vertical dashed lines mark the UKIP 2014 EU Parliament election (left) and the Brexit referendum (right). Shaded areas indicate the post-UKIP (blue) and post-Brexit (red) periods. The blue solid line corresponds to the post-UKIP period, while the red solid line corresponds to the post-Brexit period. Points represent mean attitudes, and whiskers denote 99% confidence intervals, constructed using panel-data standard errors clustered at the CSP level. Data come from the BES.

III.C. Local Environment and Belief Formation

Our conceptual framework relies on people using local xenophobic sentiment to project nationwide xenophobia. This assumes that people are aware of xenophobic sentiment in their CSP. We provide evidence in support of this argument using data from Wave 7 of BES in the context of Brexit. In the survey, households were asked the question “Most people I know will vote to remain in the European Union.” We compare share of respondents reporting yes on this question (local belief) with actual remain vote share at the CSP level. Figure 3 shows a strong positive association of 0.82 (s.e. = 0.03) between these two variables, which is statistically significant at the 1-percent level. This strong correlation suggests that people are aware of xenophobic sentiment in their local area. This means national events mainly shock beliefs over nationwide and not local xenophobia.

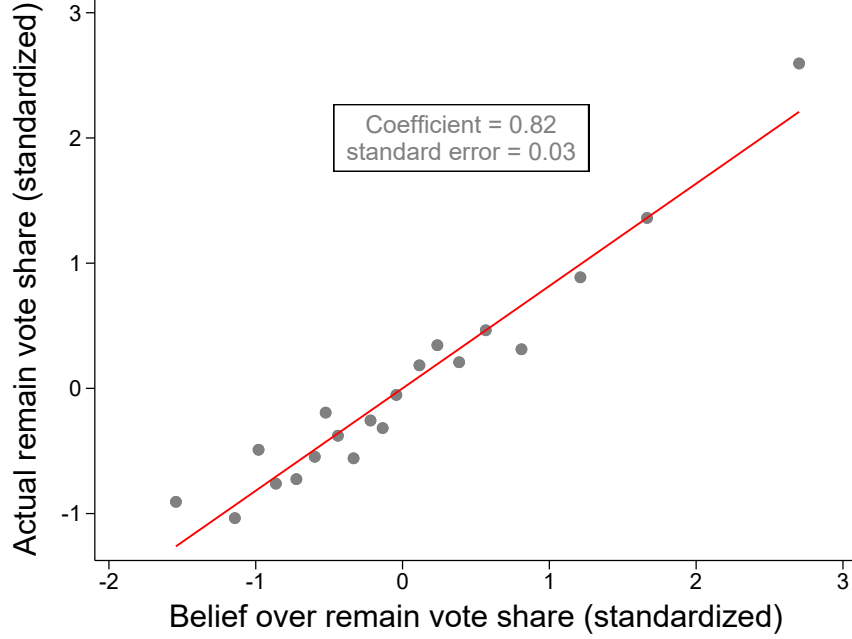


Figure 3: Remain Vote Share: Belief and Actual Outcome at Local Level

Notes. The figure shows the association at the CSP level between beliefs over remain vote share in the local area and actual remain vote share in the context of Brexit referendum. All variables are standardized to have mean 0 and standard deviation 1. Beliefs about the local remain vote share are measured using responses from Wave 7 of BES to the question, “Most people I know will vote to remain in the European Union.” Actual remain vote share data are from the UK Electoral Commission (2019).

III.D. Attitudes Toward Immigrants and Belief Shocks

Our conceptual framework argues that when national events reveal nationwide xenophobia, individuals residing in liberal areas experience belief shocks. Evidence in support of this argument warrants data on prior beliefs about the likelihood of an event happening. The BES offers data on prior beliefs for the likelihood of Brexit happening, but data on the likelihood of UKIP victory are ex post. Bearing this limitation in mind, we view results on belief shock from UKIP election as demonstrative evidence.

For the UKIP election, we use responses in Waves 4 and 5, which were conducted between 4 March 2015 and 6 May 2015, 10-12 months after the UKIP election. In the study, individuals were asked the question “Which of these parties do you think has [NO] real chance of being part of the next UK government?” UKIP was one of the listed parties. Respondents answered using a binary option, with 0 indicating “No” and 1 indicating “Yes”. We measure belief shock as 1 minus the perceived chance of UKIP winning (Yes). The average belief shock turns out to be 36.5%. Note that this measure captures belief shock after the event and only at the extensive margin, so we expect this measure to be biased downwards.

The data on beliefs about Brexit happening are from wave 8 of BES, which was conducted between 6 May 2016 and 22 June 2016, just a month before the Brexit referendum. In the survey, individuals were asked to rate the likelihood that Brexit will happen on a

scale of 0-100, where 0 implies that “UK will definitely vote to remain” and 100 implies “UK will definitely vote to leave”. Since the Leave campaign won the Brexit referendum, we measure belief shock as 100 minus the perceived likelihood of the Leave campaign winning. The average belief shock is 48.7%.

We compare belief shocks among xenophobic and liberal people living in liberal CSPs to their counterparts in xenophobic CSPs. Figure 4 presents the results. We find that xenophobic and liberal people residing in liberal areas experience a significantly larger belief shock by 0.09-0.12 standard deviations than xenophobic and liberal people residing in xenophobic areas.¹⁷

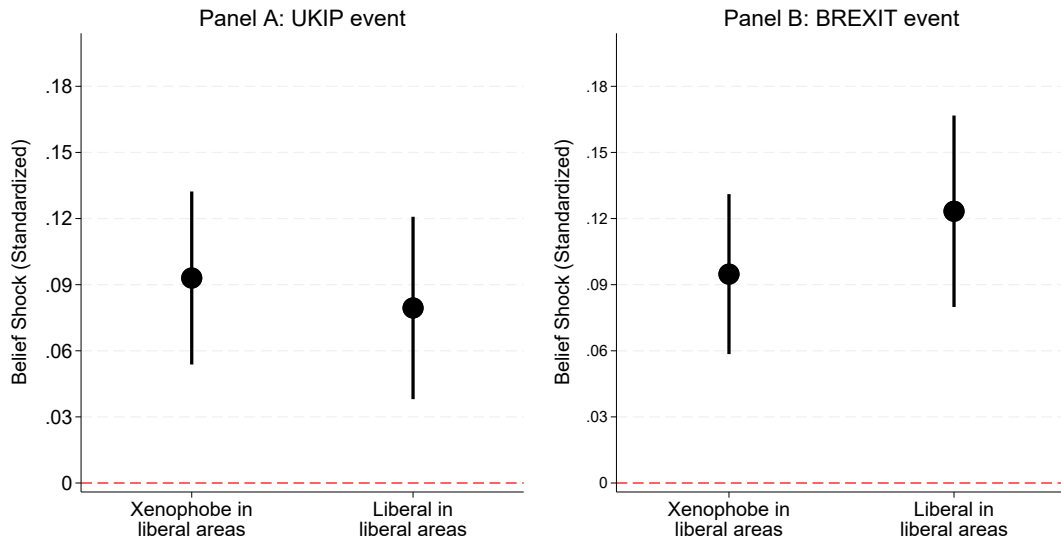


Figure 4: Belief shocks by individual attitudes and CSP type

Notes. The figure reports point estimates and 95% confidence intervals from separate regressions of standardized belief shocks on indicators for xenophobic individuals in tolerant CSPs (Panel A) and liberal individuals in tolerant CSPs (Panel B). Data are from Wave 4-5 (Panel A) and Wave 8 (Panel B) of BES.

IV. Empirical Strategy

We examine the association between hate crime and attitudes towards immigrants in post-UKIP and post-Brexit events using a difference-in-differences strategy. Our setting is similar to Korovkin and Makarin (2023), wherein every unit is treated at the same time, but there is a heterogeneous response to the treatment depending on attitudes toward immigrants. Accordingly, we compare the evolution of hate crime in CSPs with stronger versus weaker attitudes toward immigrants in response to UKIP and Brexit events. We estimate the following equation for each event separately:

¹⁷The two coefficients are not significantly different from each other either in UKIP (p -value=0.65) or Brexit (p -value=0.28)

$$Hate_{it} = \alpha + \beta(Attitudes_i \times Post-Event_t) + \gamma \mathbf{X}_{it} + \tau_t + \eta_i + \epsilon_{it} \quad (11)$$

where $hate_{it}$ is hate crime in CSP i in quarter t . *Attitudes* towards immigrants are measured on a scale of 1-7 where 1 implies weaker (negative) and 7 implies stronger (positive) attitudes toward immigrants. *Event* refers to either UKIP election to the European Parliament or Brexit referendum. For UKIP election, we use *Post-UKIP* – a binary indicator that takes a value of 1 for quarters from May 2014 onward, otherwise 0. For Brexit referendum, we use *Post-Brexit* – a binary indicator that takes a value of 1 for quarters from June 2016 onward, otherwise 0. τ_t is a fixed effect for quarter (Jan-March, April-June, July-Sept, Oct-December). It absorbs changes that affect all CSPs equally in that quarter, such as macroeconomic conditions in England and Wales. Crucially, it also absorbs seasonal variations in hate crime, which could result in higher hate crimes in summer than in winter. η_i is the fixed effect for CSP, which absorbs time-invariant differences between CSPs. ϵ_{it} is the idiosyncratic error term. Since we measure hate crime and attitudes at the CSP level, we cluster the standard errors on CSP. While conducting robustness checks, we show that our results hold when we consider spatial correlation of errors.

$\mathbf{X}$ is a vector of CSP level covariates that vary over time. Hate crime may depend on changes in the size and composition of the population in a CSP. Accordingly, we control for population size (yearly), flow of EU migrants (quarterly), as well as non-EU migrants (quarterly). This allows us to capture any policy induced changes in immigration patterns. Panels B of Table A.1 offers summary statistics on these variables. While conducting robustness checks, we consider additional variables to capture economic, social, and cultural environment in pre-event periods.

The coefficient of interest is β , which captures the effect of attitudes on hate crime in the Post-Event periods. Our conceptual framework suggests β to have a counterintuitive positive sign, that is, a steeper rise in hate crime in CSPs with more positive attitudes toward immigrants. This is because the information shock from UKIP and Brexit events is expected to be larger in such CSPs. We estimate (11) separately for both the events. For the UKIP event, we restrict the sample to periods before Brexit, that is, from Q2 2002 to Q1 2016. For the Brexit event, we use the entire sample. We might under or over estimate β in the Post-Brexit period if we fail to account for the surge in hate crime in the pre-Brexit period caused by UKIP event. So, we also present results in which we simultaneously include interaction of attitudes separately with UKIP and Brexit events.

We are worried about two concerns. First, the identifying assumption is that in the absence of the event, the evolution of hate crimes will be similar between CSPs with stronger and weaker attitudes toward immigrants. We present several strategies below to mitigate concerns under which this assumption may be violated. This includes event studies using both UKIP and Brexit, accounting for police specific factors affecting

recording of crime, and controlling for changes in economic conditions that may have led to UKIP and Brexit events (Fetzer, 2019). Second, our strategy assumes actual increase in hate crimes in the post-event periods. However, it is plausible that our results are capturing changes in the reporting behavior of victims in the post-event periods. To mitigate this concern, we use data from the Crime Survey for England and Wales (CSEW) to track changes in reporting behavior in response to the events. We also test if our results change when we adjust our measure of hate crime for under-reporting. We discuss these strategies in detail below.

IV.A. Event Studies

We conduct event studies separately for UKIP and Brexit events.

UKIP Event Study.— We examine the evolution of association between attitudes towards immigrants and hate crime before and after the UKIP event using the following equation:

$$\text{Hate}_{it} = \sum_{k=2002Q2}^{2016Q1} \beta_k \cdot T_{it}^k \times \text{Attitudes}_i + \gamma \mathbf{X}_{it} + \tau_t + \eta_i + \epsilon_{it} \quad (12)$$

where $\sum_{k=2002Q2}^{2016Q1}$ captures the time window of 48 quarters before the UKIP event (starting from Q2 2002) and eight quarters afterward (up to Q1 2016). We do not include quarters beyond this period to prevent confounding with Brexit event in Q2 2016. T_{it}^k are indicators for the event-time quarters k . $k = 0$ is Q2 2014 in which the European Parliament elections were held. The omitted category ($k = -1$) is Q1 2014. The remaining terms are the same as in Equation (11). Panel A of Figure 5 presents the results. Prior to the UKIP event, the evolution of hate crime in CSPs with weaker and stronger attitudes toward immigrants is very similar and stable over time. In contrast, in the period after the UKIP event, there is a steep rise in hate crime in areas with stronger (more positive) attitudes towards immigrants. The coefficients are individually and jointly statistically significant at the 1-percent level. This finding suggests that prior to the UKIP event, the trend in hate crime was similar across CSPs with weaker and stronger attitudes toward immigrants. However, in the post-UKIP period, the information shock led to a large surge in hate crime in CSPs with more positive attitudes toward immigrants.

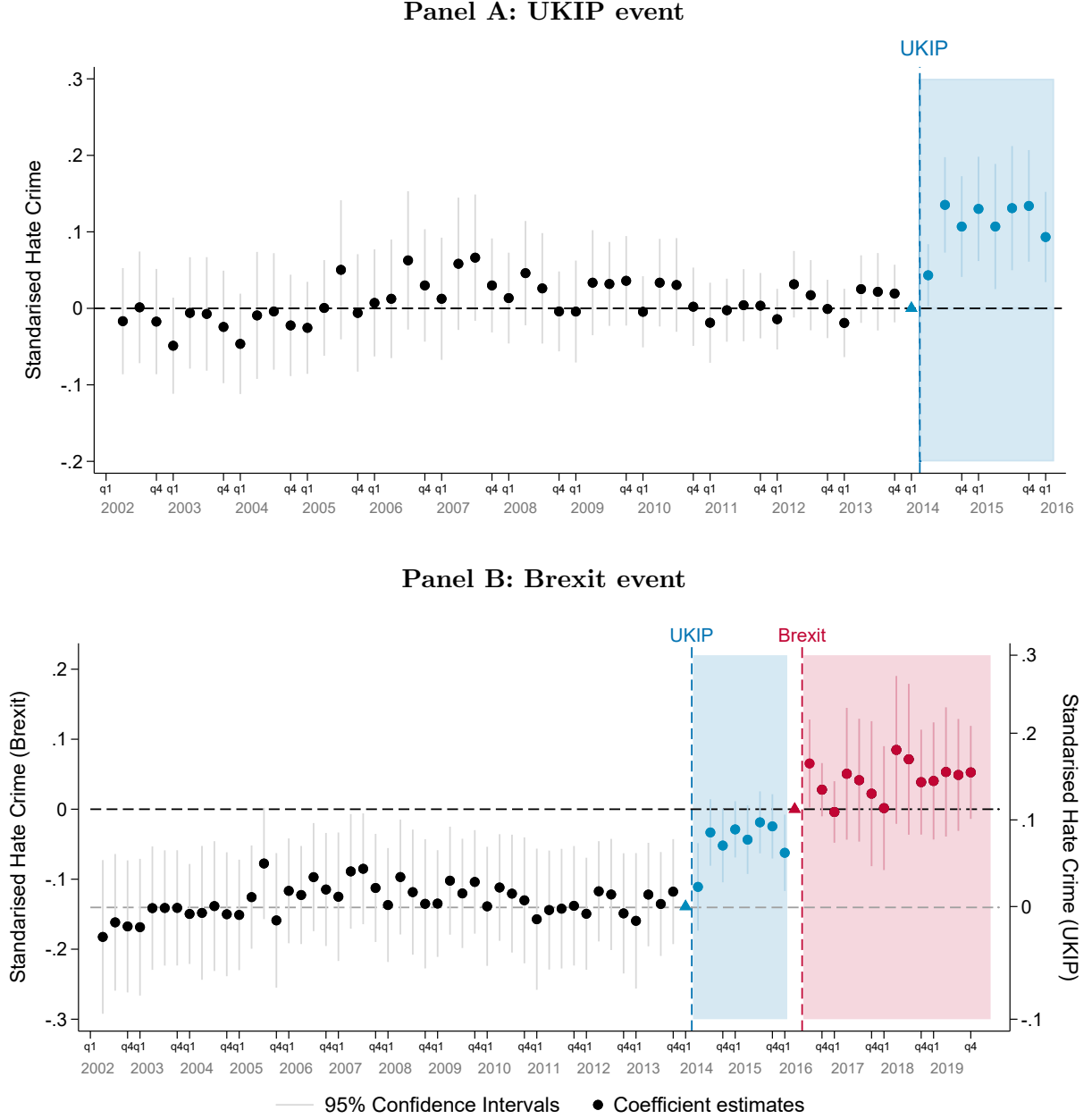


Figure 5: Event studies on the information shock of events on hate crime

Notes: Panel A plots coefficients on $T_{it}^k \times Attitudes$ from the estimation of equation 12, the UKIP event-study. The Y-axis is standardized measure of hate crime, black markers correspond to pre-UKIP period, blue markers correspond to post-UKIP event, the blue triangle corresponds to baseline (omitted) period of Q1 2014, and the blue vertical line corresponds to the UKIP election event. Panel B plots coefficients on $T_{it}^k \times Attitudes$ from the estimation of equation 13, the Brexit event-study. The primary Y-axis is standardized hate crime in all quarters that cover the Brexit event, red markers refers to post-Brexit event, the red triangle corresponds to baseline (omitted) period of Q2 2016 and the red vertical line corresponds to the Brexit referendum event. As a reference, in Panel B, we also highlight the UKIP event study using the secondary Y-axis. In both panels, in the X-axis, q1 refers to quarter 1, q4 to quarter 4, while q2 and q3 are omitted for space considerations. Ninety-five percent confidence intervals are constructed for standard errors clustered at the CSP level.

Brexit Event Study.— The second event study examines the evolution of association between attitudes toward immigrants and hate crime before and after the Brexit event

using the following equation:

$$\text{Hate}_{it} = \sum_{k=2002Q2}^{2019Q4} \beta_k \cdot T_{it}^k \times \text{Attitudes}_i + \gamma \mathbf{X}_{it} + \tau_t + \eta_i + \epsilon_{it} \quad (13)$$

where $\sum_{k=2002Q2}^{2019Q4}$ captures the time window of 57 quarters before Brexit (starting from Q2 2002) and fourteen quarters afterward (up to Q4 2019). We do not include quarters beyond this period because of COVID-19 pandemic. T_{it}^k are indicators for the event-time quarters k . Since the Brexit referendum took place on June 23, just one week before the end of Q2 2016, we choose this as the omitted category ($k = -1$), while $k = 0$ corresponds to the quarter immediately following the Brexit referendum (Q3 2016). The remaining terms are the same as in Equation (11). Panel B of Figure 5 presents the results. We find a large and statistically significant increase in hate crime in the Post-Brexit period in CSPs with more positive attitudes toward immigrants (see red solid circles). The effect peaks immediately after the Brexit referendum in Q3 2016 and remains high on average.

IV.B. Controls Variables

The event study bolsters our confidence on the absence of pre-trends in hate crime between CSPs with weaker and stronger attitudes toward immigrants. However, there could be concerns over confounding with past and concurrent changes, which we address below.

Economic Conditions.— Fetzner (2019) shows that poor economic conditions combined with austerity were responsible for the rise in UKIP popularity and support for Brexit. Therefore, one concern could be that the surge in hate crime in the post-event period is not because of information shock from UKIP or Brexit events but is actually due to the persistent effect of economic conditions from the pre-event periods. However, an empirical regularity observed across many countries is that areas with poor economic conditions tend to be associated with less rather than more tolerant attitudes toward immigrants (see for example, De Bromhead, Eichengreen and O’Rourke, 2013; Algan et al., 2017). This is also what we observe in our data, where the correlation between attitudes towards immigrants and gross disposable household income is positive and statistically significant ($r = 0.56$, p -value < 0.001). Under these circumstances, we expect CSPs with negative and not positive attitudes toward immigrants to witness a surge in hate crime in the post-event periods. Thus, it is unlikely that our estimates are confounded with pre-existing economic conditions.

Nevertheless, we adopt three strategies to address this concern. First, to the extent that changes in economic conditions occur at the regional level, we account for such changes by including region-specific time trends. Second, austerity was a result of the UK welfare reforms starting 2010. So, we control for an interaction between austerity

(financial loss per working-age adult) and post-2009 indicator. Third, we also control for an interaction of education deprivation with a post-event indicator. We measure education deprivation using the Education, Skills and Training domain of the English Indices of Deprivation (Ministry of Housing, Communities & Local Government, 2015).

A related concern could be that the surge in hate crime is the result of poor economic conditions arising from the events themselves. Notice, however, that the patterns in Figure 1 and 5 clearly show an immediate rise in hate crime in the post-event periods. This means that the concern is plausible only if economic outcomes changed immediately after the events. However, evidence suggests that changes in economic outcomes occurred with a lag of several months. In fact Bakker et al. (2022) show that consumer prices remained unchanged right after the referendum in 2016. Nonetheless, it is conceivable that the effect, especially in later periods after the event, might be in part due to post-event changes in economic conditions. So, we control for time varying gross disposable household income.

Social and Cultural Conditions.— Guriev and Papaioannou (2022) argue that several social and cultural factors can give rise to surges in populism. These include generalized trust in strangers and trust in government. We capture these variables through wave 1 and 7 of BES. For generalized trust, we use the question “Generally speaking, would you say that most people can be trusted, or that you can’t be too careful in dealing with people?” We compute the share of respondents in a CSP who choose “most people can be trusted” to measure generalized trust. For trust in government, participants are asked “How much trust do you have in Members of Parliament in general?”. They can choose their answer on a scale of 1-7, where 1 implies no trust and 7 implies a great deal of trust. We include an interact of post-event indicator with trust in strangers, as well as trust in government.

Terrorist Attacks.— One concern could be the terrorist attacks that occurred in the second quarter of 2017 in London and Manchester, which were perpetrated by immigrants and individuals of foreign descent. If these attacks led to a surge in hate crime in CSPs with more positive attitudes towards immigrants, this could result in overestimation of the effect of information shock from Brexit event. So, in all our results related to Brexit event, we control for an interaction between attitudes and post-indicator for Q2 (2017) to account for changes in hate crime that are possibly due to these terrorist attacks.

IV.C. Recording and Reporting Hate Crime

Recording Hate Crime.— Another potential concern could be changes in the recording of hate crimes in CSPs, but only to the extent that this disproportionately affected CSPs with more positive attitudes toward immigrants. We consider two strategies to deal with

this concern. First, recording practices may have changed during the pre-event periods, but the effect persists in the post-event periods. To address this concern, we control for police force-specific time trends.

Second, police recording practices may have changed in the post-event period, especially due to increased sensitivity among the police or directives to record hate crimes more accurately. To address this concern, we control for an interaction between police force and post-event indicator. This allows us to account for changes in recording that may have resulted from the events themselves.

Reporting Hate Crime.— A related concern is that the frequency of hate crimes is the same, but reporting has increased in the post-event periods. Addressing this concern fully is very difficult, so we attempt to limit its scope by using survey data from CSEW. This survey asks respondents: a) whether they have been victim of a crime in the past year; b) whether the crime has been reported to the police.¹⁸ CSEW uses these data to compute *propensity to report* different types of crimes. We focus only on those crimes that also have a hate counterpart. Table A.3 lists the average reporting rates over the study period by different types of crimes, such as wounding, assault with and without injury, arson, and vehicle damage (see Appendix for details). Note that these data are not available at the CSP level but are aggregated at England and Wales level. Nevertheless, they help us shed some light on the scope of reporting bias in three ways.

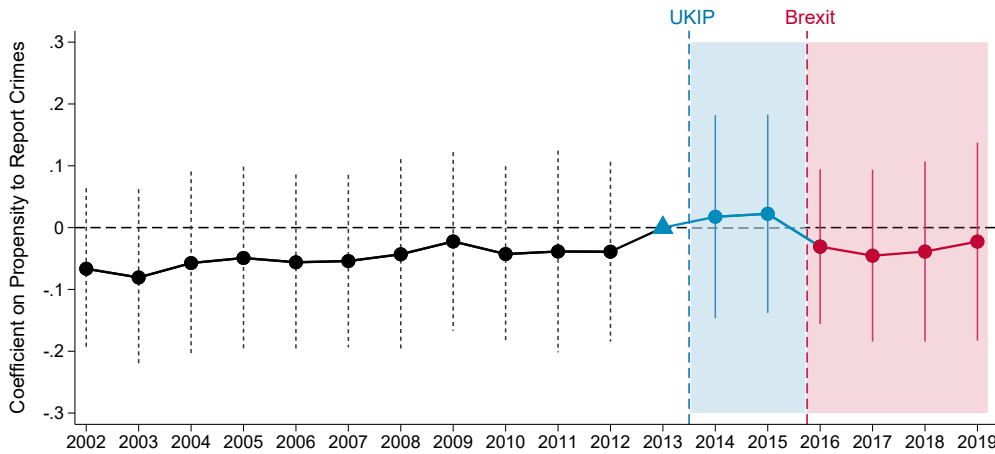


Figure 6: Propensity to Report for All Crimes

Notes. The figure shows an event-study of the average propensity to report crimes over time. The Y-axis reports estimated coefficients from a regression of the reporting rate on calendar-year indicators, relative to the baseline year (2013). The X-axis reports calendar years. Shaded areas indicate the post-UKIP (blue) and post-Brexit (red) periods. The solid blue vertical line corresponds to the post-UKIP period, while the solid red line corresponds to the post-Brexit period. Confidence intervals are at 95% level. Data are from CSEW.

¹⁸Available online at Hate crime, England and Wales, 2019 to 2020, published 13th October 2020 and also at Crime in England and Wales: Annual Trend and Demographic Tables, published 22nd July 2021.

First, we test in Figure 6 whether reporting rates have changed in response to the events. We find that the average reporting rate is flat over time and there is no difference in pre and post-event periods.

Second, we construct an adjusted measure of hate crime following a methodology that is in the spirit of Soares (2004). We exploit variation in the composition of hate crime types across CSPs. As an example, in some CSPs hate crimes are mostly due to wounding, whereas in others assaults are more common. We use crime specific reporting rates from the relevant time periods to construct an adjusted measure for each crime type and then sum these over to get an adjusted measure of hate crime. For instance, if the reporting rate of wounding crime is 50 percent in time t , we adjust hate crimes related to wounding in each CSP by factor of 2 in that time period. We then test if our results hold when we use this adjusted measure as our dependent variable.

Third, we also run our main specification separately for different types of crimes that differ in their reporting rates between 30 percent to 60 percent. The idea is that if our results are impacted by systematic under-reporting of hate crime then this should affect more those with severe under-reporting than otherwise.

IV.D. Falsification Test using Other Crimes

The information shocks from the UKIP and Brexit events are unlikely to affect crimes such as murder, burglary, and driving violations. However, there could be a surge in all of these crimes due to changes in economic conditions as well as reporting practices. Accordingly, we construct falsification tests using data on these crimes to confirm that our results are not spurious. We do not expect substitution between hate crime and other crimes unless murder, burglary, and driving violations were specifically targeted towards immigrants, which seems unlikely. Nevertheless, while conducting robustness checks we additionally control for these crimes.

V. Results

We begin by presenting our main results from the estimation of equation 11, that is, the effect of attitudes toward immigrants on hate crime in the post-event periods.

V.A. Main Result

Table 1 reports the results, which we present separately for the UKIP event in Panel A and Brexit event in Panel B. Column 1 includes fixed effects for CSPs and quarter. We find that one standard deviation increase in attitudes toward immigrants (0.458) is associated with a rise in hate crime in post-UKIP period by 0.169 standard deviations and post-Brexit period by 0.155 standard deviations. These estimates are not only economically meaningful but are also statistically significant at the 1-percent level.

Table 1: Information Shock, Attitudes, and Hate Crime

	Dependent Variable: Standardized Hate Crime			
	Quarter & CSP Fixed Effects (1)	Controls & Region $\times$ Time Trend (2)	Controls & Police $\times$ Time Trend (3)	Controls & Police $\times$ Post-Event (4)
Panel A: UKIP Event				
Attitudes $\times$ Post-UKIP	0.169 (0.024)	0.101 (0.022)	0.113 (0.021)	0.120 (0.020)
R^2	0.89	0.91	0.93	0.89
Observations	17,024	17,024	16,912	17,024
Panel B: Brexit Event				
Attitudes $\times$ Post-Brexit	0.155 (0.037)	0.126 (0.034)	0.149 (0.032)	0.161 (0.034)
R^2	0.86	0.89	0.92	0.89
Observations	21,577	21,577	21,435	21,577
Number of CSPs	304	304	302	304
CSP Fixed Effects	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Region $\times$ Quarter	No	Yes	No	No
Police $\times$ Quarter	No	No	Yes	No
Police $\times$ Post-Event	No	No	No	Yes

Notes: OLS estimates with standard errors clustered at the CSP level. FE stands for fixed effects. Control variables include population, number of EU migrants, and number of non-EU migrants. Attitudes is toward immigrants and is standardized to have mean of zero and standard deviation of 1. Panel A shows results using information shock from the UKIP event. Panel B shows results using information shock from the Brexit event. All specifications incorporate quarter and CSP fixed effects. Panel A covers a sample period from 2002 Q2 to 2016 Q1, whereas Panel B covers a sample period from 2002 Q2 to 2019 Q4. Panel B additionally includes an interaction between attitudes and an indicator for terrorist attacks (2017 Q2). The sample size is larger in Panel B because it includes fifteen additional quarters. Column (3) has fewer observations because two CSPs have singleton observations.

In column 2, we include CSP level controls like population, number of EU migrants, number of non-EU migrants, and region-specific time trends to account for regional differences in macroeconomic changes over time. The coefficient on *Attitudes $\times$ Post-UKIP* declines in magnitude to 0.101 whereas that on *Attitudes $\times$ Post-Brexit* to 0.126, but both remain statistically significant at the 1-percent level. In column 3, we introduce police force-specific time trends to account for possible changes in crime recording behavior of the police. In both panels, the coefficient increases slightly in magnitude and its standard error declines. Finally, in column 4, we control for an interaction between police force and post-event indicator to account for possible changes in crime recording behavior that may have been induced by the event itself. The coefficients increase further in magnitude and are similar in magnitude and significance to those reported in column 1.

Overall, a one-standard deviation increase in the interaction between attitudes and post-event indicator is associated with a rise in hate crime by 0.10-0.17 standard deviations.

V.B. Robustness

We carry out a number of robustness checks using the specification with our full set of controls listed in column 4 of Table 1.

Simultaneously Considering UKIP and Brexit Events.— So far, we estimate the effect of attitudes on hate crime in the post-event periods separately for the two events. We test in Table A.4 whether our results hold when we include both interaction terms simultaneously. The coefficient on *Attitudes* $\times$ *Post-UKIP* ranges from 0.071 – 0.108, and is always statistically significant. The corresponding estimate for *Attitudes* $\times$ *Post-Brexit* ranges from 0.079 – 0.090 and is also always statistically significant.

Spatial Standard Errors.— Column 1 of Table A.5 reports results after accounting for spatial correlation of standard errors using 50 km distance as cutoffs. The coefficients on both *Attitudes* $\times$ *Post-UKIP* and *Attitudes* $\times$ *Post-Brexit* remain statistically significant at the 1-percent level.

Log of Hate Crime.— Column 2 of Table A.5 reports results using log transformation of hate crime. Panel A shows that one standard deviation increase in attitudes is associated with a rise in hate crime by 7 percent in the Post-UKIP period, which is statistically significant at the 1-percent level. Panel B shows that the corresponding rise in hate crime by attitudes in the Post-Brexit period is close to 9 percent and is also statistically significant at the 1-percent level.

Economic Variables.— We test the robustness of our results to the inclusion of one by one two economic variables and then both of them simultaneously in Table 2. We include one at a time an interaction of post-2009 indicator with austerity in column 1 and an interaction of post-event indicator with an index of education deprivation in column 4. The inclusion of these controls does not lead to any major changes in the coefficient on either *Attitudes* $\times$ *Post-UKIP* or *Attitudes* $\times$ *Post-Brexit*, which remain robust in both magnitude and significance all throughout. In column 5, we include both these variables simultaneously. As before, our results hold both in magnitude and significance.

Social and Cultural Variables.— We test the robustness of our results to the inclusion of one by one two trust variables and then both of them simultaneously in Table 3. We include one at a time an interaction of post-event indicator with generalized trust in

column 1 and trust in MPs in column 2. This does not lead to any major changes in the coefficient on either $Attitudes \times Post-UKIP$ or $Attitudes \times Post-Brexit$, which remain robust in both magnitude and significance all throughout. In column 3, we include both trust variables simultaneously. As before, our results hold both in magnitude and significance.

Table 2: Information Shock, Attitudes, and Hate Crime:
Controlling for Economic Variables

	Dependent Variable: Standardized Hate Crime		
	Austerity	Education Deprivation	Both Economic Variables
	(1)	(2)	(3)
Panel A: UKIP Event			
Attitudes $\times$ Post-UKIP	0.112 (0.021)	0.130 (0.023)	0.130 (0.023)
Austerity $\times$ Post-2009	-0.058 (0.015)	– –	-0.067 (0.016)
Education $\times$ Post-UKIP	– –	0.034 (0.019)	0.067 (0.019)
R^2	0.89	0.89	0.90
Observations	17,024	17,024	17,024
Panel B: Brexit Event			
Attitudes $\times$ Post-Brexit	0.161 (0.033)	0.200 (0.036)	0.200 (0.036)
Austerity $\times$ Post-2009	-0.008 (0.019)	– –	-0.031 (0.020)
Education $\times$ Post-Brexit	– –	0.136 (0.029)	0.148 (0.031)
R^2	0.89	0.89	0.89
Observations	21,577	21,577	21,577
Number of CSPs	304	304	304
CSP Fixed Effects	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Police $\times$ Post-Event	Yes	Yes	Yes

Notes: OLS estimates with standard errors clustered at the CSP level. All columns use the baseline specification from column 4 of Table 1. Column 1 includes an interaction between post-2009 indicator and austerity. Austerity is measured as financial loss per working-age adult (Fetzer, 2019). Column 2 controls for an interaction between post-event indicator and education deprivation. We capture the latter using Education, Skills and Training domain of the English Indices of Deprivation (Ministry of Housing, Communities & Local Government, 2015). Column 3 includes both of them jointly. All variables are standardized to have mean zero and unit standard deviation.

Table 3: Information Shock, Attitudes, and Hate Crime:
Controlling for Trust Variables

	Dependent Variable: Standardized Hate Crime		
	Generalized Trust (1)	Trust in MPs (2)	Both Trust Measures (3)
Panel A: UKIP Event			
Attitudes $\times$ Post-UKIP	0.119 (0.020)	0.121 (0.021)	0.120 (0.021)
Generalized Trust $\times$ Post-UKIP	0.020 (0.047)	– –	0.021 (0.047)
Trust in MPs $\times$ Post-UKIP	– –	-0.018 (0.058)	-0.021 (0.059)
R^2	0.89	0.89	0.89
Observations	17,024	17,024	17,024
Panel B: Brexit Event			
Attitudes $\times$ Post-Brexit	0.169 (0.036)	0.157 (0.036)	0.165 (0.037)
Generalized Trust $\times$ Post-Brexit	-0.086 (0.099)	– –	-0.087 (0.099)
Trust in MPs $\times$ Post-Brexit	– –	0.073 (0.160)	0.077 (0.160)
R^2	0.89	0.89	0.89
Observations	21,577	21,577	21,577
Number of CSPs	304	304	304
CSP Fixed Effects	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Police $\times$ Post-Event	Yes	Yes	Yes

Notes: OLS estimates with standard errors clustered at the CSP level. All columns use the baseline specification from column 4 of Table 1. Column 1 controls for an interaction between post-event indicator and generalized trust, measured as the share of respondents in a CSP answering that “most people can be trusted”. Column 2 controls for an interaction between post-event indicator and trust in Members of Parliament, measured on a 1–7 scale (higher values indicate greater trust). Column 3 includes both trust measures jointly. All variables are standardized to have mean zero and unit standard deviation. Data are from BES.

Adjusted Hate Crime by Reporting Rates.— We conduct two checks. First, we show in A.7 that our results hold when we compute an adjusted measure of hate crime that takes into consideration reporting rates of different types of crime. While the coefficient on *Attitudes $\times$ Post-UKIP* is slightly smaller in magnitude than when we use the unadjusted measure, it is nonetheless statistically significant at the 1-percent level. In contrast, the coefficient on *Attitudes $\times$ Post-Brexit* is very similar in magnitude and significance to the

unadjusted measure.

Furthermore, we also run our main specification separately for different types of crimes that differ in their reporting rates. We report the results in Figure A.1 where we plot the coefficient on $Attitudes \times Post\text{-}UKIP$ in Panel A and $Attitudes \times Post\text{-}Brexit$ in Panel B obtained from a regression of different crime types as dependent variables (listed on x-axis). We find that the coefficients are largely comparable to each other and their confidence intervals overlap, thereby suggesting that under-reporting may not pose a serious concern.

Using One Attitude at a Time.— So far, our results are based on a measure of attitudes that take both economic and cultural importance of immigrants. Table A.5 shows that our results are robust to using measures of attitudes based on one of these at a time. Column 4 reports results using only the economic consideration “Do you think immigration is good or bad for Britain’s economy?”, whereas Column 5 reports results using only the cultural consideration “Do you think that immigration undermines or enriches Britain’s cultural life?”. Regardless of the question we use, the coefficients on attitudes in the post-event periods pertaining to both UKIP and Brexit are comparable in magnitude to the results presented before and retain their statistical significance at the 1-percent level.

Alternative Wave to Measure Attitudes.— So far, our results are based on attitudes measured using questions from Wave 8 for Brexit. For this event, we test whether our results also hold when we use responses to the same questions but from Wave 7, which was conducted two months before Brexit.¹⁹ Column 5 of Table A.5 reports the results. The coefficients on attitudes in the post-event periods are similar in magnitude and statistical significance to our main results.

Falsification Test using Other Crimes.— We provide further evidence in support of our results by conducting a falsification test using data on other crimes (murder, burglary, driving fines). These crimes are unlikely to be affected by information shocks from UKIP and Brexit events, but may evolve in response to other confounding changes, such as those related to recording crime or deterioration in the economic environment. Table A.8 shows the results, whereby the dependent variable is the total of other crimes, which we standardized to have a mean of zero and standard deviation of 1. Results in both panels show that the effect of attitudes on other crimes in the post-event period is always negative, which is the opposite of what we observe for hate crime. Though the coefficient is statistically significant, it is much smaller in magnitude to the one we observe in the

¹⁹We are unable to carry out this exercise for UKIP event because only one survey wave that was conducted before this event.

regression of hate crime. We further show that our results are also robust to additionally controlling for other crimes in column 2 of Table A.6.

VI. Mechanism

We argue that national events like UKIP election and Brexit referendum revealed new information on previously unknown country-wide support for xenophobia. This generated a belief shock for individuals residing in liberal CSPs. This shock was particularly instrumental for xenophobic individuals residing in liberal CSPs who realized that their attitudes align with that of the country, so they felt emboldened and resorted to hate crime.

We test this mechanism in two steps. In section III, we saw that liberal and xenophobic people residing in liberal CSPs are the ones who experience a larger belief shock. We now proceed by showing that these belief shocks map onto hate crimes. Ideally we would like to show this mapping for xenophobic individual living in tolerant areas. However, individual level data on crime are not publicly available. To circumvent this concern, we start by conducting this analysis at the CSP level. Subsequently, we also establish this link at the individual level by carrying out a proof-of-concept survey in which our outcome is not hate crime but vocalization about immigrants. We describe each of these analyses in detail below.

VI.A. Belief Shocks and Hate Crime in Post-Event Periods

We examine if belief shocks are associated with hate crime in the post-event period. Since the data on hate crime are not available at the individual level, we carry out this exercise at the CSP level. The event studies in Figure 7 reveal a strong positive association between belief shocks and hate crimes in both post-event periods.

Table 4 presents results from an econometric analysis that builds on the baseline specification in column (4) of Table 1. Column 1 uses this baseline specification, which includes CSP and quarter fixed effects, baseline controls, and police force $\times$ post-event fixed effects. It shows that a one-standard deviation increase in belief shock is associated with a rise in hate crime by 0.036 standard deviations for UKIP and 0.073 standard deviations for Brexit. In columns 2 and 3, we introduce additional sets of controls. Column 2 augments the baseline specification with the two economic interactions from column (3) of Table 2. Column 3 augments the baseline specification with the two trust interactions from column (3) of Table 3. Panel A shows that the coefficient on belief shock in the post-UKIP period remains stable at 0.036 in columns 1 and 2 and declines slightly to 0.032 in column 3, remaining statistically significant at least at the 5 percent level. In contrast, Panel B shows that the coefficient on belief shocks in the post-Brexit period is

0.073 in the baseline specification, rises to 0.098 when adding the economic interactions, and is 0.064 when adding the trust interactions; it is always statistical significant at the 1-percent level. The magnitude of these coefficients imply that one-standard deviation increase in belief shocks are associated with an increase in hate crime by 0.03-0.04 standard deviations in the post-UKIP period and between 0.06 and 0.10 standard deviations in the post-Brexit period.

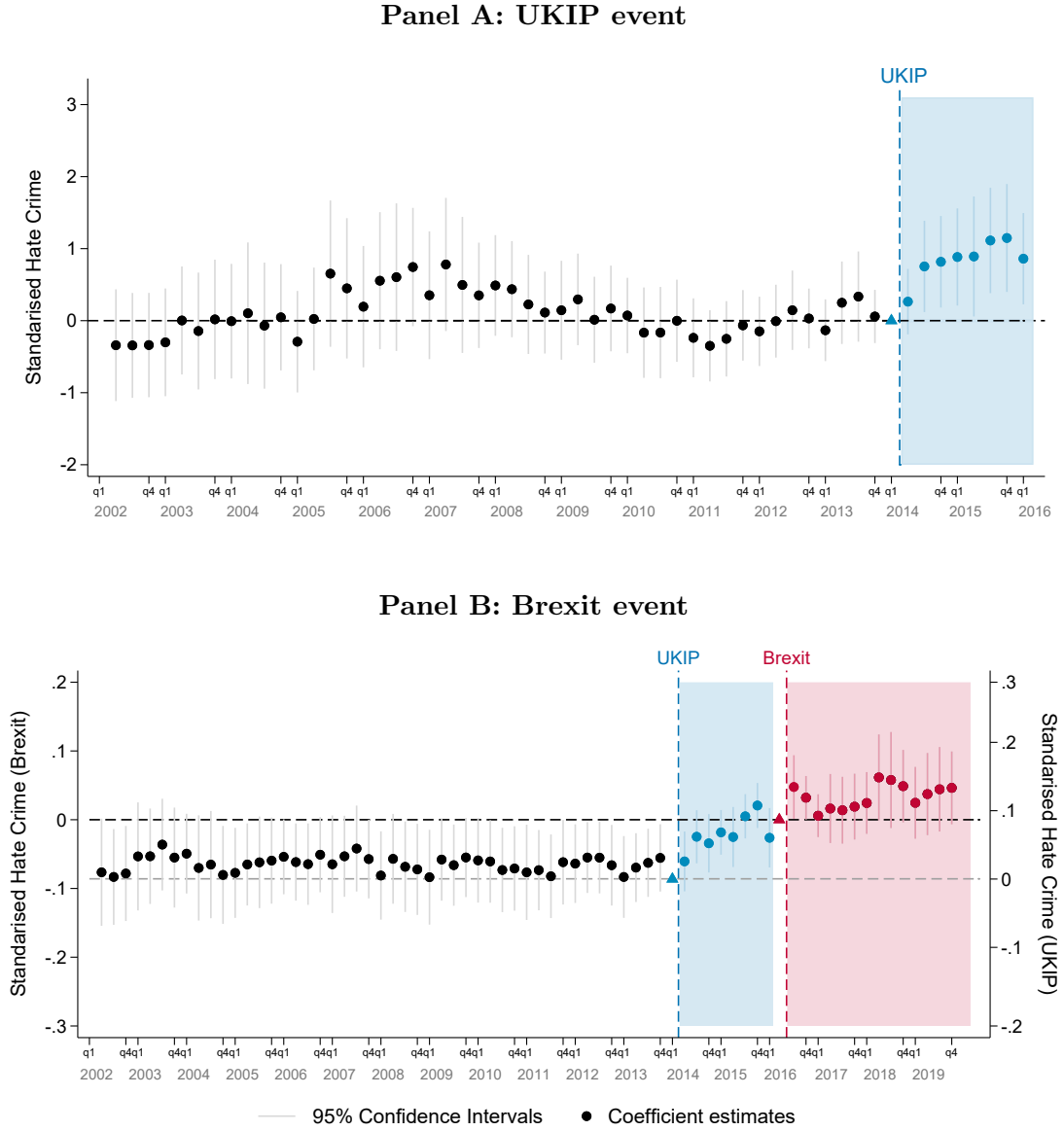


Figure 7: Event Study on Belief Shock and Hate Crime

Notes: This graph displays the result of an event-study after estimating equations 12 and 13. The coefficients are based on $Belief\ shock \times T_{it}^k$, where T_{it}^k are event-time indicators that the event happened k quarters away from UKIP and Brexit events. Black markers refer to pre-UKIP period, blue markers correspond to post-UKIP event, red marker refers to post-Brexit event, and triangles are the omitted benchmark categories for both UKIP and Brexit. The baseline period is 2014 Q1 for UKIP and 2016 Q2 for Brexit. On the X-axis, q1 refers to quarter 1, q4 to quarter 4, while q2 and q3 are omitted for space considerations. Ninety-five percent confidence intervals are constructed for standard errors clustered at the CSP level.

Table 4: Belief Shock and Hate Crime

	Dependent Variable: Standardized Hate Crime		
	Controls & Police $\times$ Post-Event	Both Economic Variables	Both Trust Measures
	(1)	(2)	(3)
Panel A: UKIP Event			
Belief Shock $\times$ Post-UKIP	0.036 (0.015)	0.036 (0.015)	0.032 (0.014)
Austerity $\times$ Post-2009	— —	-0.067 (0.016)	— —
Education $\times$ Post-UKIP	— —	0.031 (0.017)	— —
Generalized Trust $\times$ Post-UKIP	— —	— —	0.067 (0.052)
Trust in MPs $\times$ Post-UKIP	— —	— —	0.052 (0.080)
R^2	0.89	0.89	0.89
Observations	17,024	17,024	17,024
Panel B: Brexit Event			
Belief Shock $\times$ Post-Brexit	0.073 (0.028)	0.098 (0.029)	0.064 (0.028)
Austerity $\times$ Post-2009	— —	-0.030 (0.021)	— —
Education $\times$ Post-Brexit	— —	0.121 (0.029)	— —
Generalized Trust $\times$ Post-Brexit	— —	— —	0.004 (0.094)
Trust in MPs $\times$ Post-Brexit	— —	— —	0.261 (0.166)
R^2	0.88	0.88	0.88
Observations	21,577	21,577	21,577
Number of CSPs	304	304	304
CSP Fixed Effects	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Police $\times$ Post-Event	Yes	Yes	Yes

Notes: OLS estimates with standard errors clustered at the CSP level. Column 1 uses the baseline specification from column 4 of Table 1. Column 2 includes an interaction between post-2009 indicator and austerity, as well as an interaction between post-event indicator and education deprivation. Austerity is measured as financial loss per working-age adult (Fetzer, 2019) and education deprivation using Education, Skills and Training domain of the English Indices of Deprivation (Ministry of Housing, Communities & Local Government, 2015). Column 3 controls for an interaction between post-event indicator and generalized trust, as well as an interaction between post-event indicator and trust in Members of Parliament. Generalized trust is measured as the share of respondents in a CSP answering that “most people can be trusted”. Trust in Members of Parliament is measured on a 1–7 scale (higher values indicate greater trust). All variables are standardized to have mean zero and unit standard deviation.

VI.B. Attitudes, Belief Shock, and Xenophobic Behavior

We designed a survey to test for the relationship between attitude, beliefs, and tendency to engage in xenophobic behavior. The survey was run on survey platform called Prolific between February and March, 2021. Admittedly, the survey was carried out with a lag in the post-event period, so our purpose here is to offer a proof-of-concept evidence. 1448 individuals from England and Wales participated in the survey. These individuals were selected on the bases of participation in the Brexit referendum regardless of whether they supported leave or remain. We ensured that the share of participants supporting leave and remain were aligned with the actual shares in Brexit referendum, that is, 52 percent who supported leave.

In the survey, we first asked respondents to state their views on immigrants on a scale from 0 (very negative) to 10 (very positive). We invert the variable so that higher values correspond to xenophobia. Second, we asked respondents how other people in their local area view immigrants on the same scale. We keep the scale as it is so that now higher values correspond to tolerant views towards immigrants in one's own area. Finally, since we cannot ask participants their inclination to engage in hate crime, we asked them if Brexit referendum caused them to become *more or less likely* to publicly express their views on immigrants, on a scale from 1 (much less likely) to 5 (much more likely). Note that these views could be positive or negative; we did not emphasize either. We report these questions in the Appendix. In addition, the respondents also provided data on socio-economic and demographic characteristics. We list the three questions below:

1. On a scale of 0 to 10 with 0 representing very negatively and 10 representing very positively, how do you view immigrants?
2. On a scale of 0 to 10 with 0 representing very negatively and 10 representing very positively, how do you think people in your local area view immigrants?
3. Whatever your views on immigrants, did the outcome of the 2016 Brexit referendum change how comfortable you are in expressing those views? After the referendum, I became "...” to publicly express my views on immigrants.

Much more likely; more likely; neither more nor less likely; less likely; much less likely.

We use these data to identify individuals for whom the propensity to publicly express their views on immigrants changed most markedly in the post-Brexit period. For this purpose, we regress changes in public expression of views on immigrants on individual xenophobia, tolerance in one's own area, and an interaction between the two. In line with our conceptual framework, we hypothesize the interaction term to have a positive effect on the change in expression of views in the post-event period.

Table 5 shows that this is indeed the case. The coefficient on the interaction term is 0.197 and is statistically significant at the 1-percent level. This result implies that xenophobic individuals living in tolerant areas are more likely to increase their expression of views on immigrants after the Brexit referendum than those living in less liberal areas; the later also enters with a positive coefficient but is much smaller in magnitude and is also statistically insignificant.

Table 5: Changes in Public Expression of Views on Immigrants in the Post-Event Period

	Dependent variable: Change in Expression	
	Without controls (1)	With controls (2)
Individual Xenophobia	0.065 (0.054)	0.048 (0.054)
Tolerant Area	-0.044 (0.047)	-0.045 (0.047)
Individual Xenophobia $\times$ Tolerant Area	0.197 (0.052)	0.193 (0.052)
R^2	0.057	0.067
<i>Observations</i>	1,448	1,448
Controls	No	Yes

Notes: OLS estimates with standard errors clustered at the individual level. The dependent variable “Change in expression” is toward immigrants and measures the change in how likely respondents are to express their opinions on immigrants following the Brexit referendum. Controls include individual age, gender, education level, income and employment. All variables are standardized with mean zero and standard deviation of one.

Together, these results indicate that xenophobic and liberal people in tolerant areas are the ones who experienced belief shock in the post-event period. Moreover, at the CSP level, these belief shocks correspond to larger surges in hate crime in the post-event periods. Finally, xenophobic individuals from tolerant areas are much more likely to express their view on immigrants publicly in the aftermath of the events.

VII. Conclusion

In this paper we study a paradoxical result from information shocks in fueling xenophobic behavior. We find that national revelations of anti-immigrant sentiment via UKIP’s 2014 election to the European Parliament and the 2016 Brexit referendum resulted in a surge in hate crimes in communities with historically positive attitudes toward immigrants. Using a difference-in-differences strategy, we show a strong positive and statistically significant effect of attitudes towards immigrants on hate crimes in the post-event periods. Event

study plots suggest that these results are unlikely to be because of the violation of the parallel trends assumption. Moreover, our results are robust to including a wide variety of economic variables as well as their interaction with post-event indicator.

Our conceptual framework, rooted in heterogeneous priors and strategic complementarities, suggests that these results arise because xenophobic minorities in tolerant areas update their beliefs once they realize that their attitudes and that of the nation are aligned. This diminishes the social cost of committing hate crimes and normalizes xenophobic behavior. In line with our conceptual framework, we use data from BES to demonstrate that people with anti-immigrant attitudes in liberal areas experience a larger belief shock than people with anti-immigrant attitudes in xenophobic areas. Using data on the perceived likelihood of the events, we construct measures of belief shock and show that it has a large positive and significant effect on hate crimes in the post-event periods. A proof-of-concept survey reveals that anti-immigrant individuals from liberal areas are much more likely to voice their concerns in the post-event periods.

These findings challenge conventional narratives, highlighting how populist shocks erode pluralistic ignorance not in places where xenophobia is entrenched but also in liberal areas, potentially amplifying societal divisions. This has important implications for policymakers aiming to foster social cohesion and address hate crimes.

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ONLINE APPENDIX:
Attitudes toward Immigrants,
Information Shocks, and Hate Crime

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Appendix A Theory

Proof Proposition 1 As discussed in the text, standard Bayesian updating delivers,

$$E(\mu | P^d, \bar{\mu}) = \frac{\Theta}{1 + \Theta} P^d + \frac{1}{1 + \Theta} \bar{\mu}. \quad (14)$$

Consider a generic linear symmetric equilibrium where, for all i , $a_i = k\alpha_i + \gamma P_i^d + (1 - k - \gamma) \bar{\mu}$, and, hence, $\bar{a} = (k + \gamma) \mu + (1 - k - \gamma) \bar{\mu}$ and $\bar{a}^d = (k + \gamma) P^d + (1 - k - \gamma) \bar{\mu}$. The optimal action for individual j living in district d is

$$\begin{aligned} a_j &= \theta \alpha_j + (1 - \theta) [\lambda E_j(\bar{a} | P^d, \bar{\mu}) + (1 - \lambda) E_j(\bar{a}^d | P^d, \bar{\mu})] \\ &= \theta \alpha_j + (1 - \theta) (1 - k - \gamma) + (1 - \theta) (k + \gamma) [\lambda E_j(\mu | P^d, \bar{\mu}) + (1 - \lambda) P^d] \end{aligned}$$

Substituting for $E_j(\mu | P^d, \bar{\mu})$ we obtain the following system

$$\begin{aligned} k &= \theta \\ \gamma &= (1 - \theta) (k + \gamma) \left(\lambda \frac{\Theta}{1 + \Theta} + 1 - \lambda \right) \end{aligned}$$

which, after substituting $k = \theta$ delivers $\gamma = \frac{\theta(1-\theta)(\Theta+1-\lambda)}{\theta+\lambda(1-\theta)+\theta\Theta}$ as described in proposition 1.
QED

Proposition 2 *For sufficiently large σ , the difference in hate crime rates before and after the event is an increasing function of P^d .*

Proof of Proposition 2 A second order Taylor approximation of our standard normal distribution delivers¹

$$\begin{aligned} f(\tilde{a}_0^d) - f(\tilde{a}_1^d) &\cong f(\tilde{a}_0^d) (\tilde{a}_1^d - \tilde{a}_0^d) \left[\tilde{a}_0^d - \frac{1}{2} \left((\tilde{a}_0^d)^2 - 1 \right) (\tilde{a}_1^d - \tilde{a}_0^d) \right] \\ &= f(\tilde{a}_0^d) \frac{\bar{a}_0^d - \bar{a}_1^d}{\theta^2 \sigma} \left[\mathbf{a} - \bar{a}_0^d + \frac{1}{2} \left(\frac{(\mathbf{a} - \bar{a}_0^d)^2}{\theta^2 \sigma} - 1 \right) (\bar{a}_1^d - \bar{a}_0^d) \right] \end{aligned}$$

Substituting in (10) we get

$$\frac{f(\tilde{a}_0^d)}{\theta \sqrt{\sigma}} [\hat{\varkappa}(\theta + \gamma_1) + \gamma_0 - \gamma_1] \quad (15)$$

where $\hat{\varkappa} \equiv \frac{\bar{a}_1^d - \bar{a}_0^d}{\theta^2 \sigma} \left[\mathbf{a} - \bar{a}_0^d + \frac{1}{2} \left(\frac{(\mathbf{a} - \bar{a}_0^d)^2}{\theta^2 \sigma} - 1 \right) (\bar{a}_1^d - \bar{a}_0^d) \right]$. Recall that $\bar{a}_0^d$, $\bar{a}_1^d$, γ_1 and γ_0 are independent of σ . If $\sigma \rightarrow \infty$, $\hat{\varkappa} \rightarrow 0$ and hence we can be certain that the expression in the square bracket in (15) is positive (since $\gamma_0 > \gamma_1$). By continuity, this implies that there exists a value $\tilde{\sigma}$ such that expression (15) is unambiguously positive when $\sigma > \tilde{\sigma}$. QED

¹Recall that, for the standard normal distribution, we have $f'(x) = -xf(x)$ and $f''(x) = (x^2 - 1)f(x)$.

Appendix B Empirical Strategy and Results

Summary Statistics.— Table A.1 reports summary statistics over hate crime, attitudes toward immigrants, and additional variables that we use while conducting robustness checks.

Summary Statistics	
Panel A: Main Variables	
Hate Crime	31.330 (42.778)
Attitudes Toward Immigrants (Wave 1)	3.433 (0.480)
Attitudes Toward Immigrants (Wave 8)	3.636 (0.458)
Panel B: Control Variables	
Population (1000s)	180.304 (118.067)
Number of EU migrants	271.197 (452.551)
Number of Non-EU migrants	177.304 (329.212)

Notes: The table reports the mean and standard deviations in parentheses. All variables are measured at the level of CSPs in England and Wales. The sample covers 304 CSPs from 2002 Q1–2019 Q4. Hate Crime is the average count of police-recorded hate offenses over 71 quarters. Attitudes toward immigrants (W1) is the average of attitudes relevant for UKIP event analysis, whereas that for W8 is for Brexit event analysis. These data are from BES. Population is the ONS mid-year estimate and is expressed in thousands. Number of EU migrants and Non-EU migrants captures migrant flows measured using quarterly National Insurance Number (NINO) registrations, split into EU and non-EU nationals.

Validation Exercise.— A.2 validates our measure of attitudes by showing that these measures are, as expected, negatively correlated with vote share for both UKIP and Leave campaign of Brexit.

Table A.2: Attitudes Toward Immigrants and Political Outcomes:
Validation Exercise using UKIP and Leave Vote Share

	Dependent variable: Vote Share	
	UKIP (1)	Brexit (2)
Attitudes Toward Immigrants	-0.368 (0.051)	-0.577 (0.049)
R^2	0.69	0.76
<i>Observations</i>	304	304
Controls	Yes	Yes

Notes: OLS estimates with robust standard errors. Column (1) uses UKIP vote share as the dependent variable; column (2) uses support for leaving the EU (Brexit). Controls include average age, gender, education level, income, ethnicity and religion. All variables are standardized with mean zero and standard deviation of one.

Reporting Rates by Crime Category.— We match 12 different types of hate crime listed in ONS to five types of crime in CSEW. Table A.3 provides the details. Column 1 lists five crime types in the CSEW that also have hate equivalent. Column 2 lists corresponding hate crimes in ONS. Column 3 lists reporting rates from CSEW that are averaged over 18 years. While computing adjusted crime statistics, we use year specific weights. It is based on Table D10 of “*Crime in England and Wales: Annual Trend and Demographic Tables*”.

Table A.3: Crime Equivalency and Reporting Rates

CSEW Offense Group	Mapping and Reporting Parameters	
	Hate Crime in ONS (Racially or Religiously Aggravated)	Average Reporting Rate (%)
Wounding	• Inflicting grievous bodily harm without intent • Actual bodily harm and other injury • Assault with injury	59.8
Assault with minor injury	• Less serious wounding	39.5
Violence without injury	• Assault without injury • Harassment • Public fear, alarm or distress	38.7
Arson and other criminal damage	• Criminal damage to a building other than a dwelling • Criminal damage to a dwelling • Other criminal damage • Criminal damage	42.5
Criminal damage to a vehicle	• Criminal damage to a vehicle	28.5

Notes: Column 1 lists five crime types that have hate equivalents in CSEW. Column 2 lists corresponding hate crimes in ONS. Reporting rates are average over 18 years and expressed as percentages.

Simultaneously Considering UKIP and Brexit Events.— Table A.4 presents results after considering simultaneously the interaction of attitudes toward immigrants with post-UKIP and post-Brexit events. Our results hold in both magnitude and significance.

Table A.4: Attitudes Toward Immigrants and Hate Crime:
Considering UKIP and Brexit Simultaneously

	Dependent Variable: Standardized Hate Crime		
	Region × Time Trend	Police × Time Trend	Police × Post-Event
	(1)	(2)	(3)
Attitudes × Post-UKIP	0.071 (0.022)	0.087 (0.021)	0.108 (0.019)
Attitudes × Post-Brexit	0.079 (0.031)	0.089 (0.029)	0.090 (0.033)
R^2	0.89	0.92	0.89
Observations	21,577	21,435	21,577
Number of CSPs	304	302	304
CSP Fixed Effects	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Region × Time Trend	Yes	No	No
Police × Time Trend	No	Yes	No
Police × Post-Event	No	No	Yes

Notes: OLS estimates with standard errors clustered at the CSP level. All columns include controls for CSP population size and the number of EU and non-EU migrants. Additionally we control for terrorist attacks by interacting attitudes with the post-event indicator (2017 Q2). Fixed effects include CSP and quarter fixed effects. Column (2) has fewer observations because two CSPs have singleton observations.

Additional Robustness Checks.— Table A.5 shows that our results are robust to a variety of additional checks that we implement using our final specification from column 4 in Table 1. This includes using spatial standard errors over 50 km distance (column 1), log of hate crime (column 2), consider economic (“immigration is good for the economy”) and cultural (“immigration enriches cultural life”) attitudes separately (column 3-4), and when we use attitudes toward immigrants from Wave 7 of BES (column 5).

Table A.5: Attitudes Toward Immigrants and Hate Crime:
Robustness Checks

	Dependent Variable: Standardized Hate Crime				
	Spatial Errors (1)	Hate in Logs (2)	Attitudes (Economy) (3)	Attitudes (Culture) (4)	Attitudes (Wave 7) (5)
Panel A: UKIP Event					
Attitudes $\times$ Post-UKIP	0.120 (0.029)	0.069 (0.013)	0.115 (0.020)	0.117 (0.020)	— —
R^2	0.15	0.86	0.90	0.90	—
Observations	17,024	17,024	17,024	17,024	—
Panel B: Brexit Event					
Attitudes $\times$ Post-Brexit	0.161 (0.033)	0.086 (0.018)	0.159 (0.035)	0.161 (0.031)	0.142 (0.032)
R^2	0.14	0.87	0.89	0.89	0.89
Observations	21,577	21,577	21,577	21,577	21,577
Number of CSPs	304	304	304	304	304
CSP Fixed Effects	Yes	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Police $\times$ Post-Event	Yes	Yes	Yes	Yes	Yes

Notes: OLS estimates with standard errors clustered at the CSP level in parentheses except in column 1 which reports standard errors adjusted for spatial correlation of errors in 50 km distance. Column 2 reports results using log of hate crime, columns 3-4 using separately the economic and cultural component of attitudes, and column 5 using measures of attitudes toward immigrants from Wave 7 of BES. All columns include controls for CSP population size and the number of EU and non-EU migrants. Panel B additionally includes an interaction between attitudes and an indicator for terrorist attacks (2017 Q2). Fixed effects include CSP, quarter, and police force $\times$ post-event fixed effects.

Additional Controls.— Table A.6 shows that our results hold when we control for an interaction between lifting of migration restrictions for new EU member states like Romania and Bulgaria and post-event indicator in column 1. Our results are also robust to controlling for other crimes that are unrelated to hate.

Table A.6: Attitudes Toward Immigrants and Hate Crime:
Controlling for Other Crimes

	Dependent Variable: Standardized Hate Crime
	Panel A: UKIP Event
Attitudes $\times$ Post-UKIP	0.120 (0.020)
R^2	0.89
Observations	17,024
	Panel B: Brexit Event
Attitudes $\times$ Post-Brexit	0.158 (0.034)
R^2	0.89
Observations	21,577
Number of CSPs	304
CSP Fixed Effects	Yes
Quarter Fixed Effects	Yes
Controls	Yes
Police $\times$ Post-Event	Yes
Other Crimes	Yes

Notes: OLS estimates with standard errors clustered at the CSP level. The specification controls for other crimes, defined as the aggregation of murder, burglary, and driving fines. It includes controls for CSP population size and the number of EU and non-EU migrants. Panel B additionally includes an interaction between attitudes and an indicator for terrorist attacks (2017 Q2). Fixed effects include CSP, quarter, and police force $\times$ post-event fixed effects.

Reporting Hate Crime.— Table A.7 reports results using adjusted hate crime as the dependent variable. Our results are comparable in magnitude and significance to those obtained using the unadjusted hate crime measure. Furthermore, Figure A.1 shows that our results also hold when we use different types of crime as our dependent variable.

Table A.7: Attitudes Toward Immigrants and Hate Crime:
Adjusting for Under-Reporting of Hate Crime

	Dependent Variable: Standardized Hate Crime
	Panel A: UKIP Event
Attitudes $\times$ Post-UKIP	0.070 (0.016)
R^2	0.90
Observations	17,024
	Panel B: Brexit Event
Attitudes $\times$ Post-Brexit	0.146 (0.034)
R^2	0.87
Observations	21,577
Number of CSPs	304
CSP Fixed Effects	Yes
Quarter Fixed Effects	Yes
Controls	Yes
Police $\times$ Post-Event	Yes

Notes: OLS estimates with standard errors clustered at the CSP level. The dependent variable is standardized hate crime adjusted for under-reporting using reporting-rate parameters derived from the CSEW. The regression includes controls for CSP population size and the number of EU and non-EU migrants. Panel B additionally includes an interaction between attitudes and an indicator for terrorist attacks (2017 Q2). Fixed effects include CSP, quarter, and police force $\times$ post-event fixed effects.

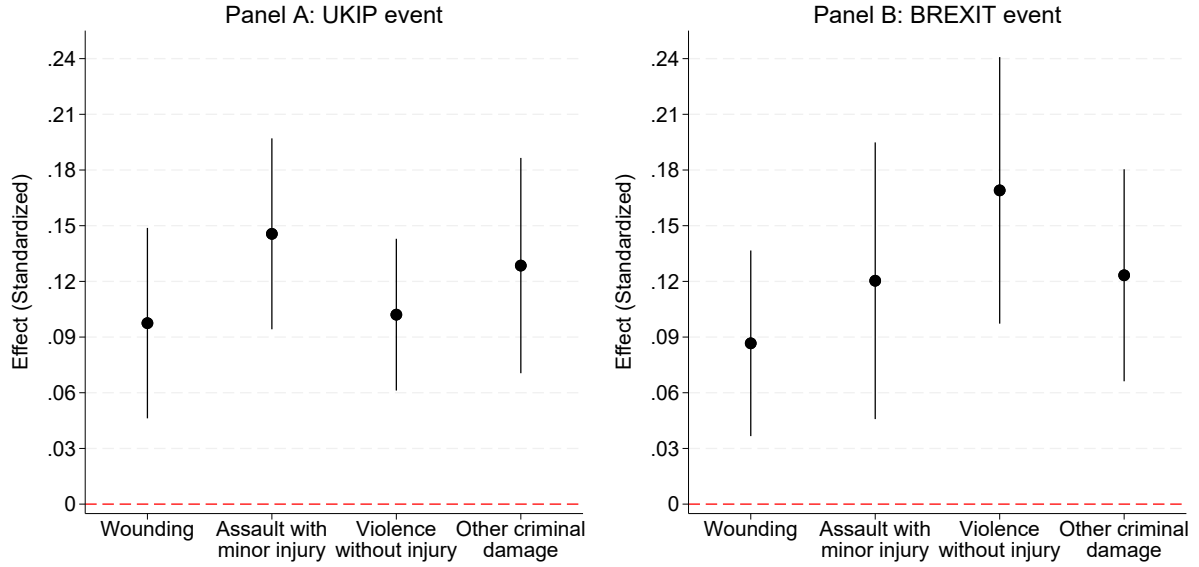


Figure A.1: Information Shocks and Hate Crime by Crime Type

Notes. The figure reports point estimates and 95% confidence intervals from separate regressions of standardized hate crime outcomes on a post-event indicator. Panel A corresponds to the UKIP electoral shock, estimated using Wave 1 data, while Panel B corresponds to the Brexit referendum shock, estimated using Wave 8 data. Crime outcomes are grouped into four categories: wounding; assault with minor injury; violence without injury; and other criminal damage. All specifications include controls for CSP population size and the number of EU and non-EU migrants. Panel B additionally includes a control for terrorist attacks interacted with the post-event indicator (2017 Q2). Fixed effects include CSP, quarter, and police force $\times$ post-event fixed effects. Standard errors are clustered at the CSP level.

Falsification Test.— Table A.8 reports results from a falsification test in which we use other crimes like burglary, murder, and driving related crimes as the dependent variable.

Table A.8: Attitudes Toward Immigrants and Other Crimes:
Falsification Test

	Dependent Variable: Other Crimes
	Panel A: UKIP Event
Attitudes $\times$ Post-UKIP	-0.048 (0.016)
R^2	0.90
Observations	17,024
	Panel B: Brexit Event
Attitudes $\times$ Post-Brexit	-0.042 (0.017)
R^2	0.90
Observations	21,577
Number of CSPs	304
CSP Fixed Effects	Yes
Quarter Fixed Effects	Yes
Controls	Yes
Police $\times$ Post-Event	Yes

Notes: OLS estimates with standard errors clustered at the CSP level. The dependent variable is standardized *other crimes*, defined as the aggregation of murder, burglary, and driving fines. The regression includes controls for CSP population size and the number of EU and non-EU migrants. Panel B additionally includes an interaction between attitudes and an indicator for terrorist attacks (2017 Q2). Fixed effects include CSP, quarter, and police force $\times$ post-event fixed effects.